IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Executive Summary

- Methodology
  - **The REST API and web scraping** were used for data collection about Falcon 9 first-stage landing by SpaceX, and then **data wrangling** was conducted.
  - Then, **exploratory data analysis (EDA)** was performed using **visualization and SQL**.
  - **Interactive visual analytics** was done with **maps and a dashboard**.
  - Finally, **predictive analysis using classification models** was performed about success/failure of the Falcon 9 first-stage landing.
- Conclusion
  - **The development of Falcon 9 for its first-stage landing progressed well**, which is good news for potential customers to reduce the launch cost.
  - To predict whether the Falcon 9 first-stage lands successfully or not, a machine learning model of **logistic regression** is recommended.
  - The launch cost reduction by the reusable Falcon 9 first-stage is very positive to space exploration and utilization, but **how data science will help you** in winning the space race depends on whether you are **a customer or competitor of SpaceX in the market**.

Introduction

- Project Background and Context
  - **SpaceX** advertises that its launcher, **Falcon 9**, will be cheaper to launch payloads to the Earth's orbit than other launch services because **the first stage** of the launcher can be **landed and reused**, instead of being disposed.
  - The cost of a launch is **62 million USD**, according to the SpaceX website, versus the maximum of 165 million USD of other launch service providers.
  - Thus, if potential customers and competitors can know whether the first stage of Falcon 9 lands successfully or not, it is helpful to **determine the cost of a launch and bid against SpaceX**.
- Problems to Find Answers
  - **In what conditions Falcon 9 first stage successfully lands**, specifically:
    - ➢ Launch site, payload mass, orbit type, customer, booster type, and landing site
  - **Yearly trend of the success in the landing**
  - **Machine learning model to predict the success**

Section 1

Methodology

Methodology (Summary)

- In this project, a programming language **Python** was used for each stage of the analysis.
- Collect data: the **REST API and web scraping** (Wikipedia)
- Perform **data wrangling**
  - ➢ Missing values and data types identified
  - ➢ Number of launches from each site calculated
  - ➢ Number and occurrence to each orbit and landing outcome calculated
  - ➢ Landing outcome label created
- Perform **exploratory data analysis (EDA)** using **visualization and SQL**
- Perform **interactive visual analytics** using **Folium and Plotly Dash**
- Perform **predictive analysis** using **classification models**
  - ➢ A feature selected as outcome
  - ➢ Data preprocessed (other features standardized) as conditions of landing success/failure
  - ➢ Dataset split into training and test data
  - ➢ Models selected and trained (logistic regression, support vector machine, decision tree, and K-nearest neighbors)
  - ➢ Models evaluated

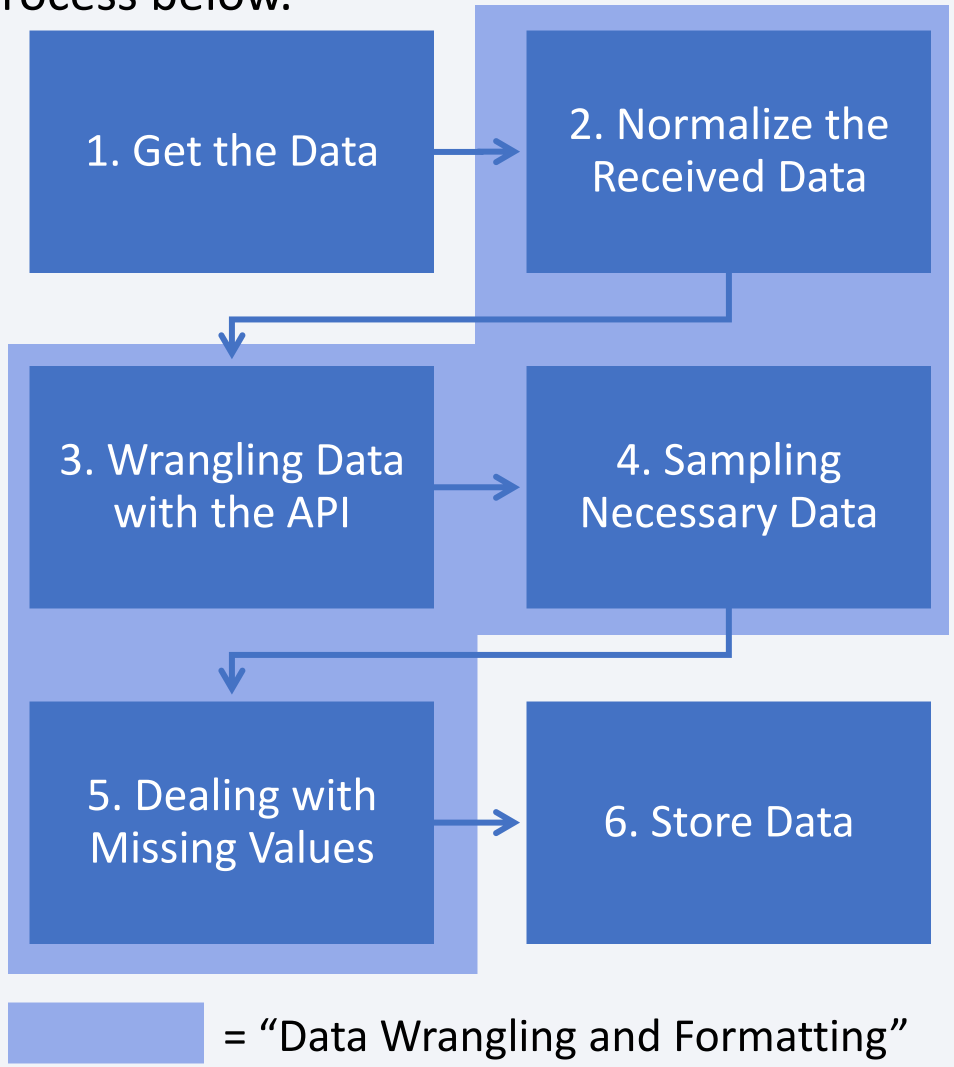
Data Collection

- Data sets were collected as follows.
  - **The SpaceX REST API**: information about SpaceX launches was gathered by calling the SpaceX API with its endpoint at https://api.spacexdata.com/v4/, using the request method GET. The obtained JSON data was converted to a dataframe, parsed for required fields, and stored to a CSV file after data wrangling and formatting.
  - **A Wikipedia page**: Falcon 9 historical launch records were extracted from a Wikipedia page at the target URL of https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches&oldid=1027686922, using the GET method to send an HTTP request. The obtained HTML table was parsed and data extracted using BeautifulSoup, then converted into a dataframe. The data was exported to a CSV file after data wrangling and formatting.
- The next three slides describe **the detailed processes of data collection** with key phrases and in flowcharts.

Data Collection – SpaceX API (1/3)

Information about SpaceX launchers was gathered with the process below.

1. Get the Data:
   - Send an API request by request.get( ) to the API endpoint at https://api.spacexdata.com/v4/launches/past
   - Check the received API response by print(response.content)
2. Normalize the Received Data:
   - Convert the extracted JSON data to a Pandas dataframe by pd.json_normalize(response.json( ))
3. Wrangling Data with the API (Continued to the next slide):
   - Get information about the launchers from IDs given for each launch in the dataframe above, using the columns of rocket, payloads, launchpad and cores

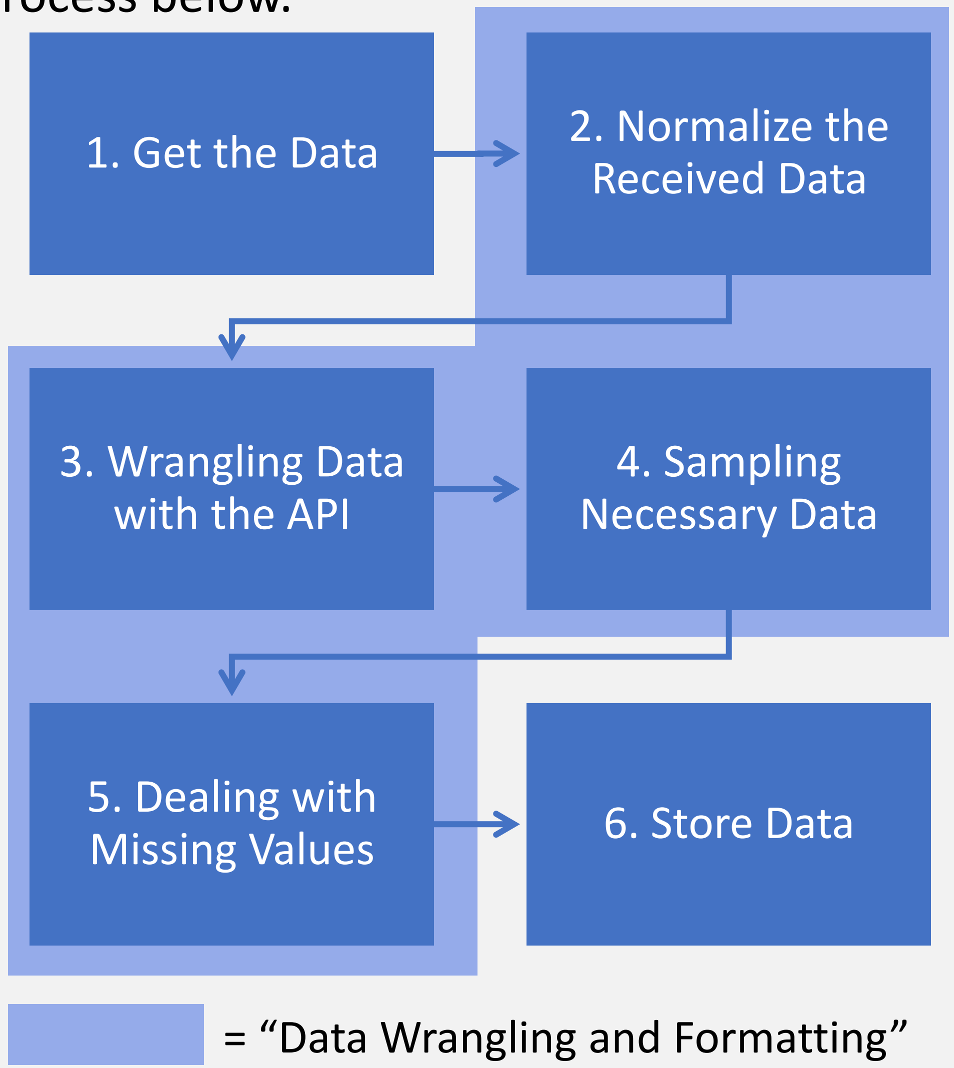


Data Collection – SpaceX API (2/3)

Information about SpaceX launchers was gathered with the process below.

3. Wrangling Data with the API (Continued from the previous slide):
   - For the purpose, send API requests again with getBoosterVersion( ) for https://api.spacexdata.com/v4/rocket; getLaunchSite( ) for https://api.spacexdata.com/v4/launchpad; getPaylodData( ) for https://api.spacexdata.com/v4/payload; and getCoreData( ) for https://api.spacexdata.com/v4/cores which includes the data such as outcomes of the landing

4. Sampling Necessary Data:
   - Combine the columns in 3. above into a Python dictionary and create a Pandas dataframe from the dictionary
   - Filter the dataframe to only include Falcon 9 launchers in the BoosterVersion column by deleting Falcon 1 data, and save the filtered data to a new Pandas dataframe and reset the FlightNumber column in the data

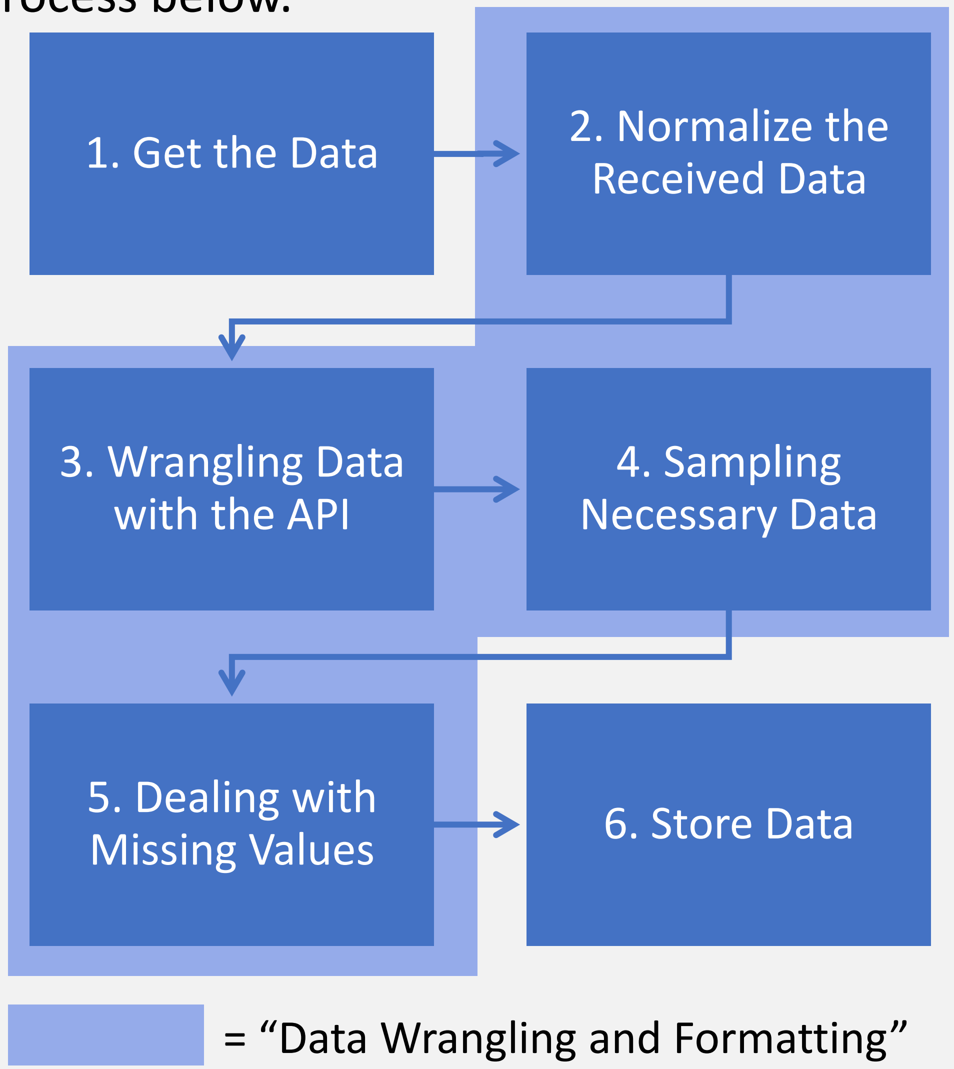


Data Collection – SpaceX API (3/3)

Information about SpaceX launchers was gathered with the process below.

5. Dealing with Missing Values:
   - The new dataframe contains five Nulls in the PaylaodMass column (need for treatment) and 26 in the LandingPad column (no need for treatment because no landing pad was used in those launches)
   - Calculate the mean for the payload mass using .mean( ) and replace the missing values by .replace({'PayloadMass':{np.nan:#calculated mean#}})
6. Store Data:
   - Export the data to a CSV file

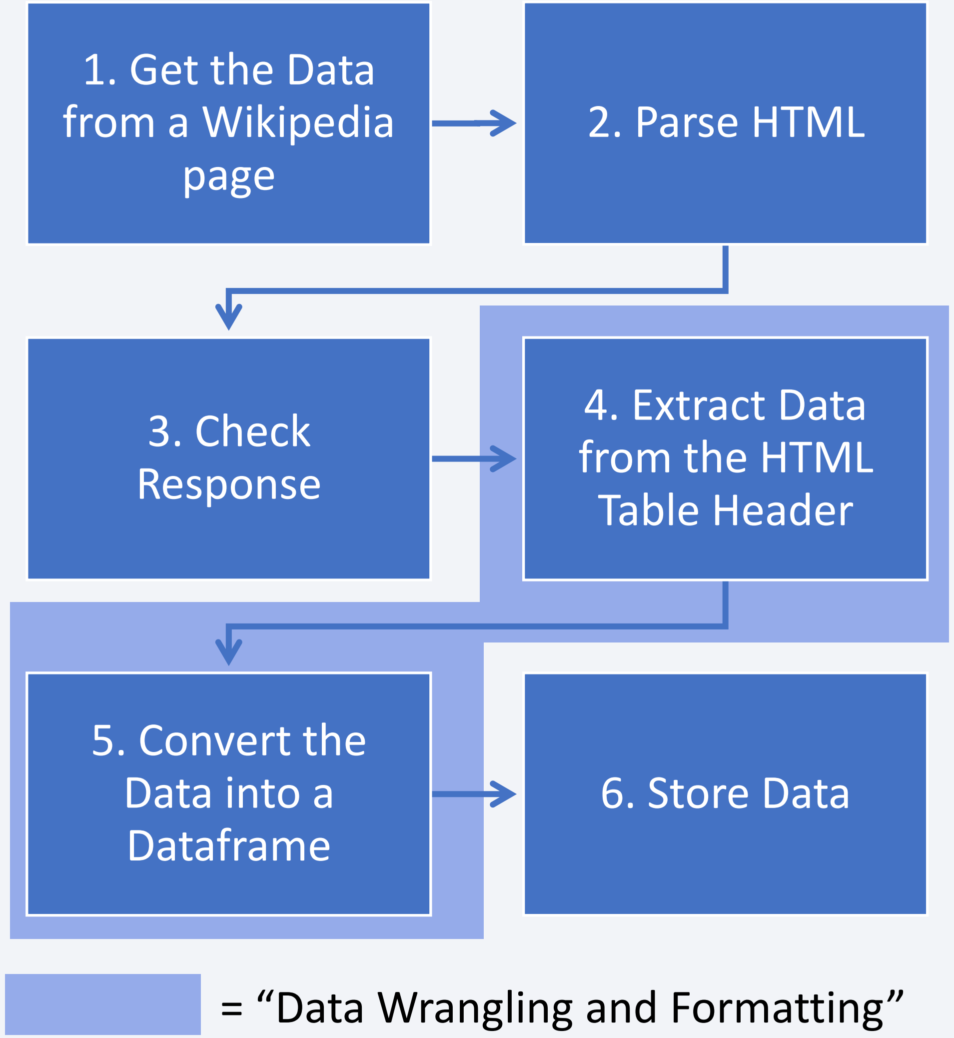
Find the Jupyter Notebook file for the data collection using the SpaceX REST API at https://github.com/J-Cym-Burke/biz-analytics/blob/main/jupyter-labs-spacex-data-collection-api.ipynb.



Data Collection – Web Scraping (1/3)

Falcon 9 historical launch records was gathered with the process below.

1. Get the Data from a Wikipedia page:
   - Set a target URL at https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches&oldid=1027686922, and send an HTTP request by resuests.get( ) to fetch the Falcon 9 launch HTML page
2. Parse HTML:
   - Create a BeautifulSoup object from the HTML response using BeautifulSoup(response.text, ‘html.parser’)
3. Check Response:
   - Verify if the object was created properly by print( )



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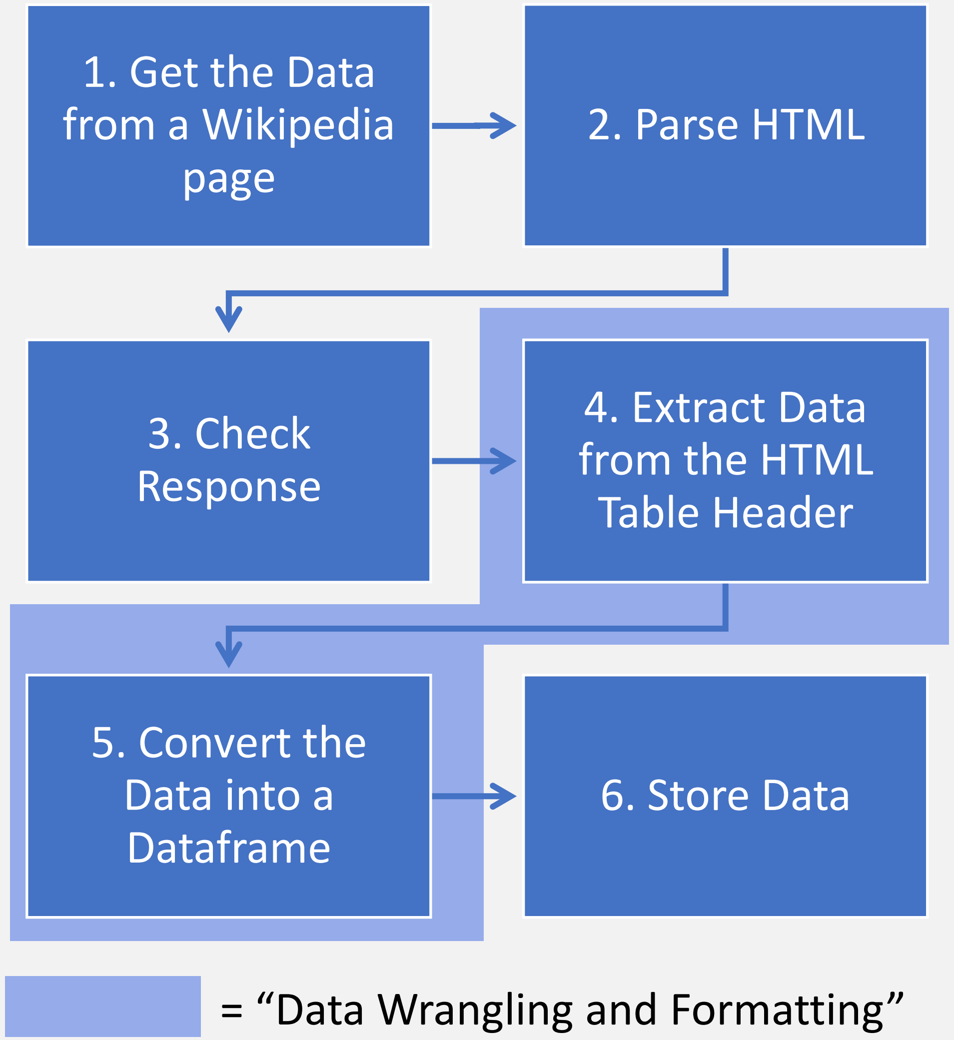
Data Collection – Web Scraping (2/3)

Falcon 9 historical launch records was gathered with the process below.

4. Extract Data from the HTML Table Header:
    - Find all tables on the webpage by soup.find_all('table')
    - Iterate through the <th> elements and apply a helper function that returns the landing status from the HTML table cells
    - Check the extracted column names by print( )
5. Convert the Data into a Dataframe:
    - Create a Pandas dataframe by parsing the launch HTML table as follows;
        - ➢ First, create an empty Python dictionary with keys from the extracted column names by dict.fromkeys( )
        - ➢ Next, remove irrelevant columns by del and initial the dictionary with each value to be an empty list and add some new columns
        - ➢ Third, fill up the dictionary with launch records extracted from table rows
        - ➢ Finally, create a dataframe



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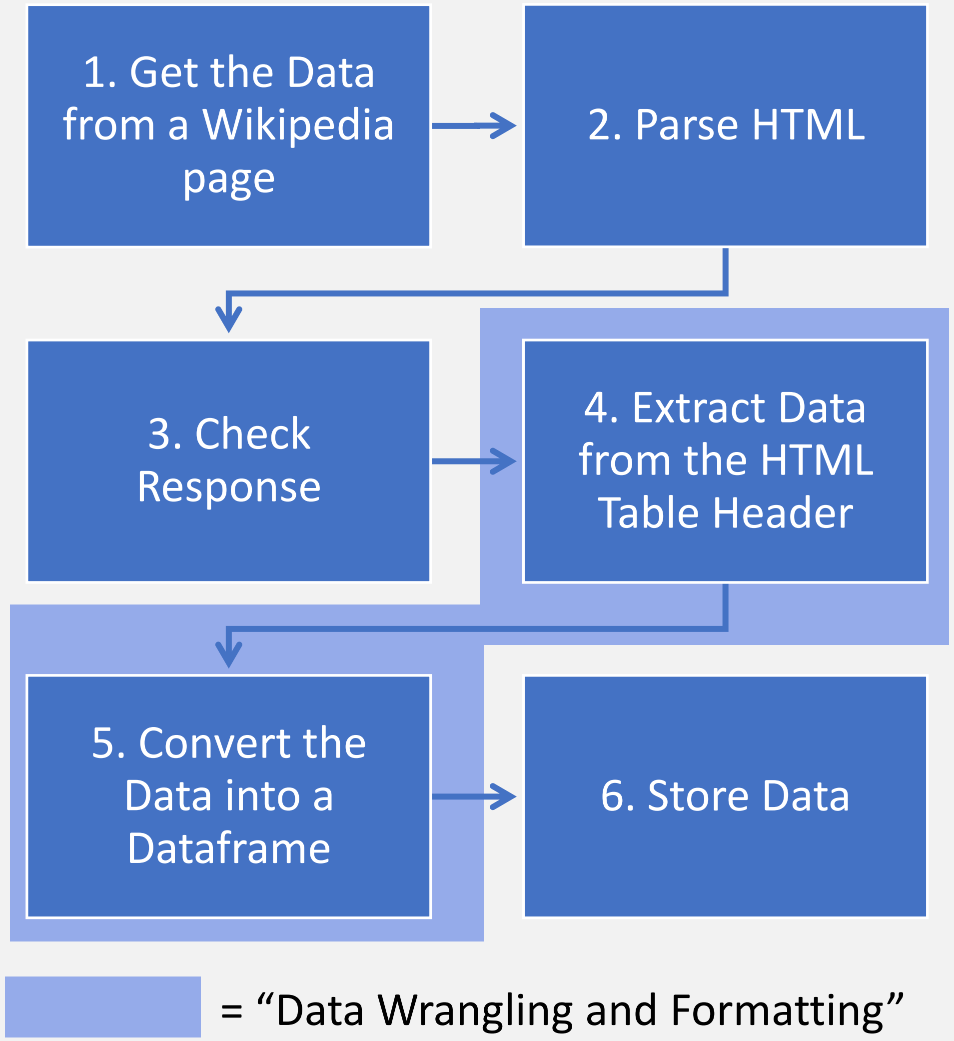
Data Collection – Web Scraping (3/3)

Falcon 9 historical launch records was gathered with the process below.

6. Store Data:
    - Export the data to a CSV file

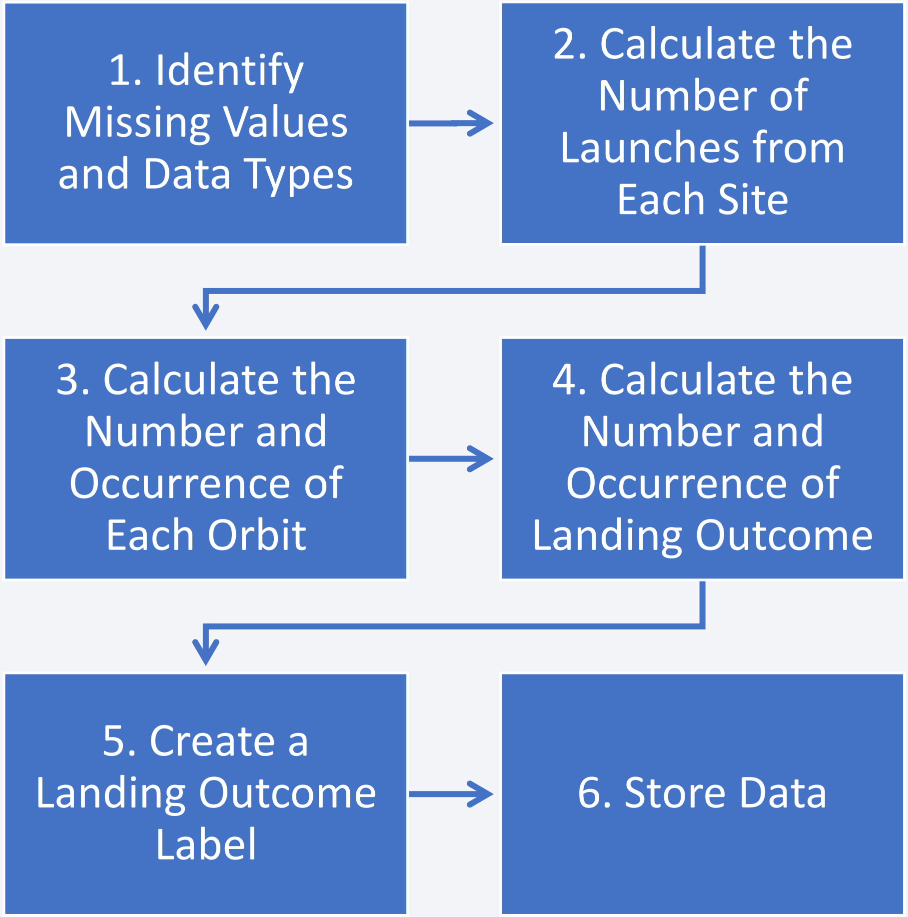
Find the Jupyter Notebook file for the data collection using web scraping at https://github.com/J-Cym-Burke/biz-analytics/blob/main/jupyter-labs-webscraping.ipynb.



Data Wrangling (1/2)

The collected data was wrangled in the process below.

1. Identify Missing Values and Data Types:
   - After loading the SpaceX dataset by pd.read_csv( ), the percentage of missing values in each attribute was calculated by df.isnull( ).sum( )/len(df)*100
   - Then, data types of each column were identified by df.dtypes
2. Calculate the Number of Launches from Each Site:
3. Calculate the Number and Occurrence of Each Orbit:
4. Calculate the Number and Occurrence of Landing Outcome:
   - The above numbers and occurrences were calculated using .value_counts( )
   - See Appendix 1 for the results

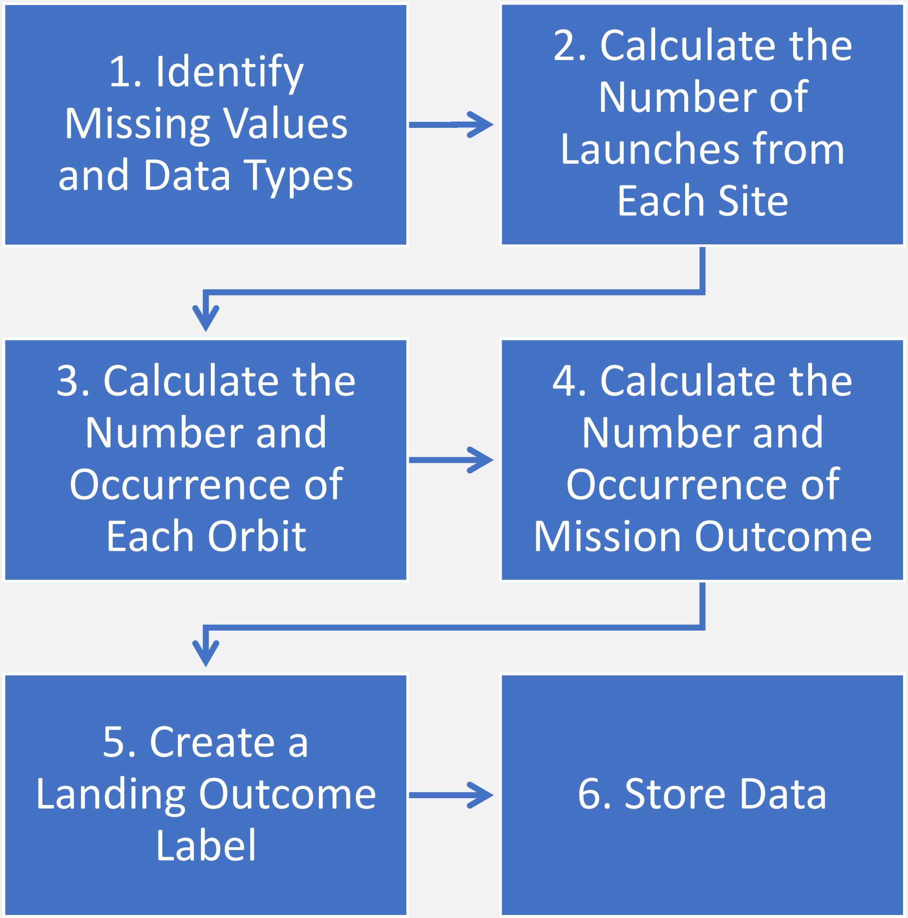


Data Wrangling (2/2)

The collected data was wrangled in the process below.

5. Create a Landing Outcome Label:
   - After obtained the number and occurrence of landing outcomes, a set of unsuccessful outcomes was created using iteration and set( )
   - A list was created in which the value is 0 if the landing outcome is in the unsuccessful set (failure) or 1 otherwise (success), then, the list was assigned to a classification variable of the launch outcomes
   - The variable was converted into a feature ‘Class’ in a Pandas dataframe
   - Finally, the success rate was calculated using .mean( ), which returned about 66.7%
6. Store Data:
   - The data was exported to a CSV file

Find the Jupyter Notebook file for the data wrangling at https://github.com/J-Cym-Burke/biz-analytics/blob/main/labs-jupyter-spacex-Data%20wrangling-v2.ipynb.



EDA with Data Visualization

- In this project, an exploratory data analysis (EDA) was conducted to develop **a preliminary understanding of which variables affect the success of Falcon 9 first-stage landing**.

- For the purpose, 5 scatter plot charts, 1 bar chart, and 1 line chart were created:
  - ➢ 3 scatter plot charts to see the relationship between flight numbers and each of the payload mass, launch sites, and orbit type
  - ➢ 2 scatter plot charts to see the relationship between payload mass and each of the launch sites and orbit type
  - ➢ 1 bar chart to see the relationship between success rate and orbit type
  - ➢ 1 line chart to see the yearly trend of the average success rate

Find the Jupyter Notebook file for the EDA with data visualization at https://github.com/J-Cym-Burke/biz-analytics/blob/main/jupyter-labs-eda-dataviz-v2.ipynb.

EDA with SQL (1/2)

- After loading the SQL extension of Python, connecting to the database using sqlight, and then converting and loading a Pandas dataframe (SpaceX dataset originally in a CSV format) to a table for Db2 database, SQL queries were performed to check the following information.
- Launch sites as unique names by the query SELECT * FROM *;
- 5 launches from the site starting with 'KSC' by SELECT * FROM * WHERE * LIKE '*%' LIMIT *;
- Total payload mass carried by boosters launched by NASA Commercial Resupply Service (CRS) by SELECT *, SUM(*) FROM * WHERE * = '*';
- Average payload mass launched using booster F9 version 1.1 by SELECT '*', AVG('*') FROM * WHERE '*' = '*';
- Date on which the first successful landing on a drone ship was achieved by SELECT '*', MIN(*) FROM * WHERE '*' = '*';
- Booster names which successfully landed on ground pads and had payload mass greater than 4,000kg and less than 6,000kg by SELECT '*' FROM * WHERE '*' = '*' AND '*' BETWEEN * AND *;

EDA with SQL (2/2)

- Total number of success and failure in mission outcomes by SELECT TRIM('*') AS *, COUNT(*) AS * FROM * GROUP BY TRIM('*');
- Booster names which carried the maximum payload mass by SELECT '*' AS '*' FROM * WHERE '*' = (SELECT MAX('*') FROM *);
- Successful landing on a ground pad in each month of 2017 with booster versions and launch sites by SELECT SUBSTR(*) AS *, *, '*', '*', '*' FROM * WHERE '*' = '*' AND SUBSTR(*)='*';
- Count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between 2010-06-04 and 2017-03-20 in descending order by SELECT '*', COUNT(*) AS * FROM * WHERE * BETWEEN '*' AND '*' GROUP BY '*' ORDER BY * DESC;

Find the Jupyter Notebook file for the EDA with SQL at https://github.com/J-Cym-Burke/biz-analytics/blob/main/jupyter-labs-eda-sql-edx-sqllite-v2.ipynb.

Build Interactive Maps with Folium

- To find some **geographical patterns about launch sites**, the following 3 interactive maps were created with a Python library Folium.
- First, to overview **the relative location** of each launch site from the equator and coastline, the sites' locations and names were added using folium.Circle based on their latitudes and longitudes and folium.Marker for popup labels.
- Second, to see which launch sites had **higher success rates in the landing** than others, color-labeled markers were added by creating a MarkerCluster object from markers of landing records (success or failure) and then adding a folium.Marker to the cluster.
- Finally, to explore and analyze **the proximities of each launch site**, a MousePosition was added to show the coordinates when pointing a mouse on the map and to calculate the distance between the two coordinates, then was created a folium.Marker to show the distance. In addition, lines were drawn from a launch site to each point of the closest coastline, city, railway, and highway using folium.PolyLine.

Find the Jupyter Notebook file for the interactive maps with Folium at https://github.com/J-Cym-Burke/biz-analytics/blob/main/lab-jupyter-launch-site-location-v2.ipynb. Use nbviewer to see the interactive maps because GitHub represents a Jupyter Notebook in the static HTML and interactive elements will not work as in the original file.

Build a Data Visualization Dashboard with Plotly Dash

- To **obtain insights** on the success rates and numbers per launch site, payload range, and booster version **visually**, a dropdown menu and a few charts were created using Python Plotly Dash framework.
- **A dropdown component** was added with labels and values **of all of the launch sites and each of them**, i.e., CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, and CCAFS SLC-40 (see Appendix 2 for details of each site including the names in fully spelled form).
- **A slider** was added to get charts based on **a range of payload mass**.
- **A pie chart** was added to show **the success rate** by **launch site/all sites and payload mass**.
- **A scatter point chart** was added to show **the success rate** by **launch site, payload mass, and booster version**.

Find the Jupyter Notebook file for the dashboard with Plotly Dash at https://github.com/J-Cym-Burke/biz-analytics/blob/main/lab-jupyter-interactive-dashboard-as-replacement-of-SkillsNetwork.ipynb. Use nbviewer to recreate the dashboard as in the original file because GitHub represents a Jupyter Notebook in the static HTML and interactive elements will not work as it is.

Predictive Analysis (Classification) (1/3)

The predictive analysis on the Falcon 9 first-stage landing was conducted in the process below.

1. Loading the Dataset:
   - Load the dataset in a CSV format and convert it to a Pandas dataframe
2. Selecting Feature:
   - Choose a feature ‘Class’ and create a Numpy array by to_numpy( ), assigning it to Y
3. Preprocessing the Data:
   - Standardize other features in the data as X by preprocessing.StandardScaler( ) and reassign them to X by transform.fit_transform( )

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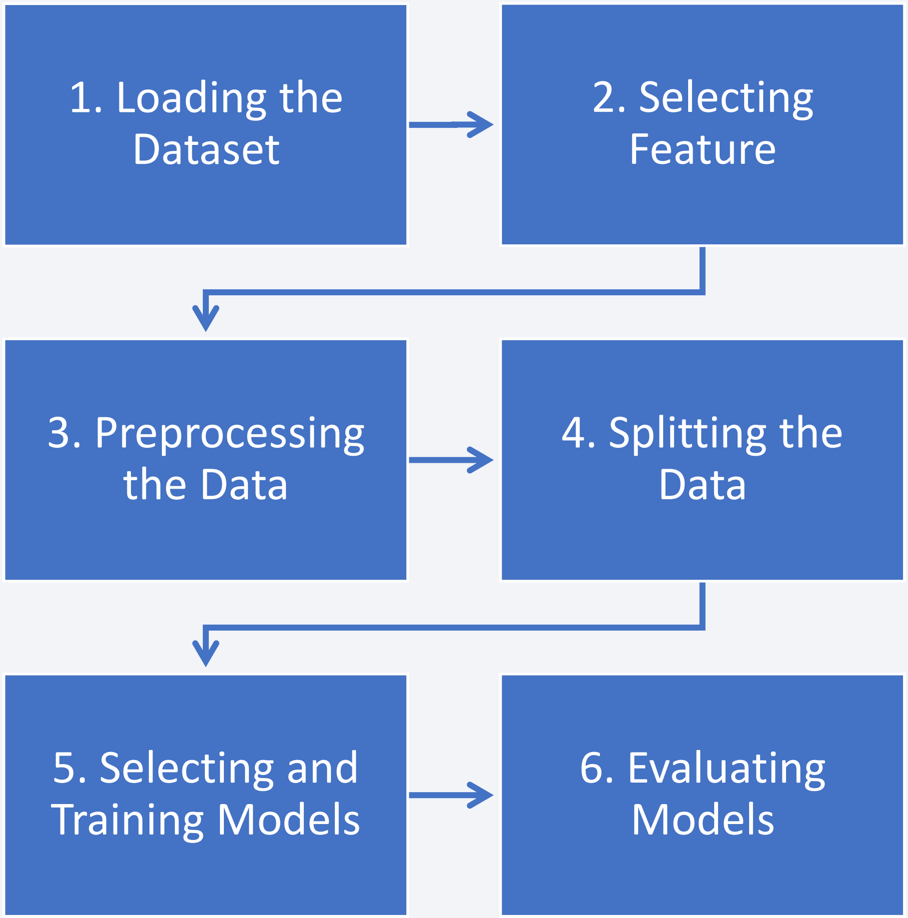

Predictive Analysis (Classification) (2/3)

The predictive analysis was conducted in the process below.

4. Splitting the Data:
   - Split the data into training and test data by train_test_split( ) with the parameter test_size of 0.2 and random_state of 2.
5. Selecting and Training Models:
   - For each of logistic regression, support vector machine, decision tree, and k-nearest neighbors, create Python objects and then GridSearchCV objects by GridSearchCV( ) with cv = 10, and fit the object to find the hyperparameters by fit.( ).
   - Display the tuned hyperparameters by .best_params_ and the accuracy on the validation data by .best_score_.
   - Calculate the accuracy on the test data by .score( ) and create a confusion matrix by plot_confusion_matrix( ).

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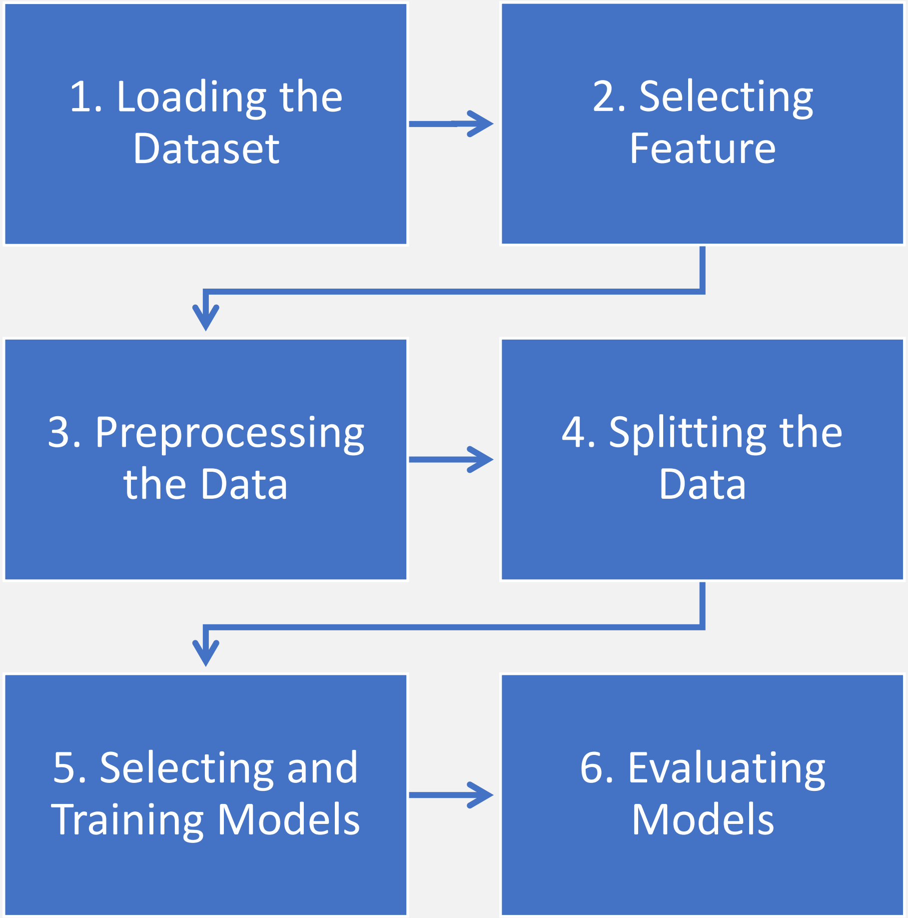
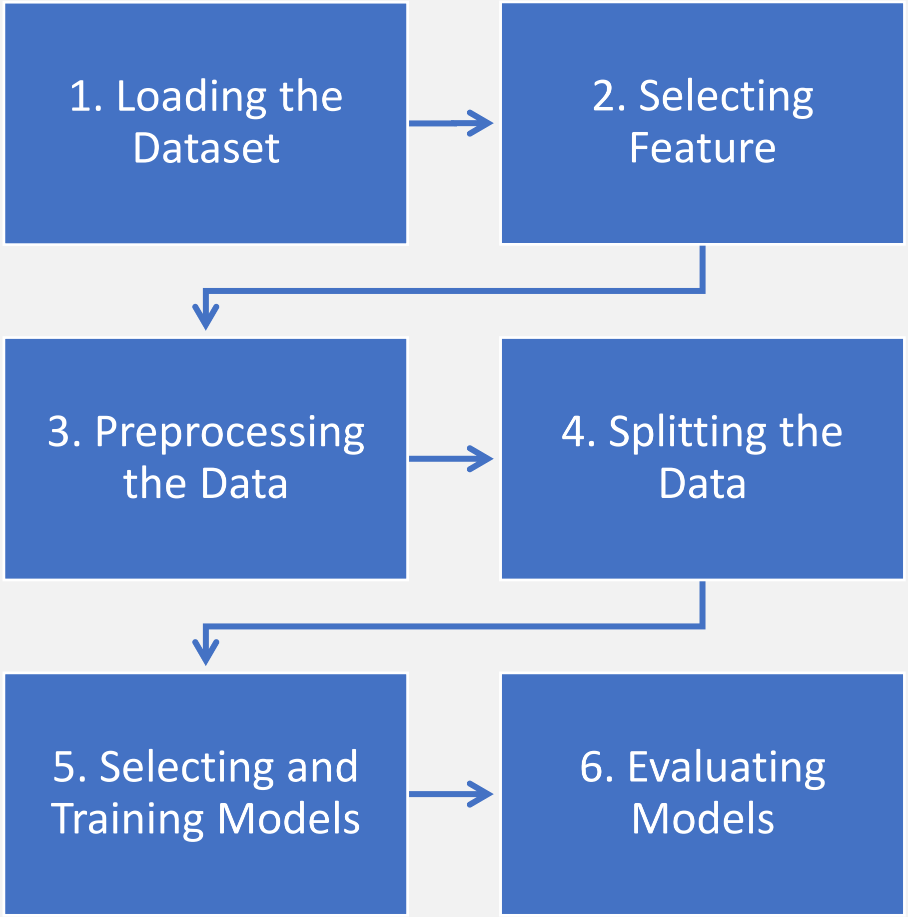


Predictive Analysis (Classification) (3/3)

The predictive analysis on the Falcon 9 first-stage landing was conducted in the process below.

6. Evaluating Models:
   - Find the model that performs best by calculating and comparing metrics in accuracy, which measures overall correctness of the models on the previous slide.

Find the Jupyter Notebook file for the predictive analysis at https://github.com/J-Cym-Burke/biz-analytics/blob/main/SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb.



Results

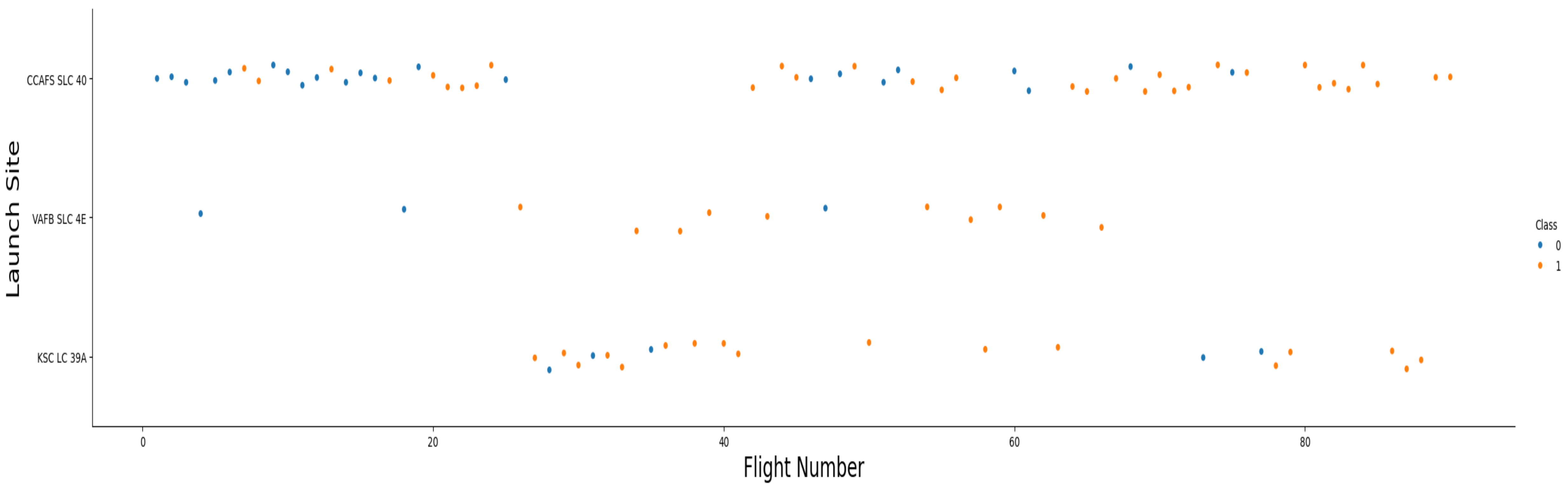
- Detailed results from the methods described so far will be explained in the following sections.
  - Section 2: Insights Drawn from EDA
  - Section 3: Launch Sites Proximities Analysis (with interactive analytics demo in screenshots)
  - Section 4: Build a Dashboard with Plotly Dash (same as above)
  - Section 5: Predictive Analysis (Classification)

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site (1/2)

The scatter point chart below shows the relationship between flight number and launch site. Orange dots are success in the Falcon 9 first-stage landing whereas blue ones are failures.



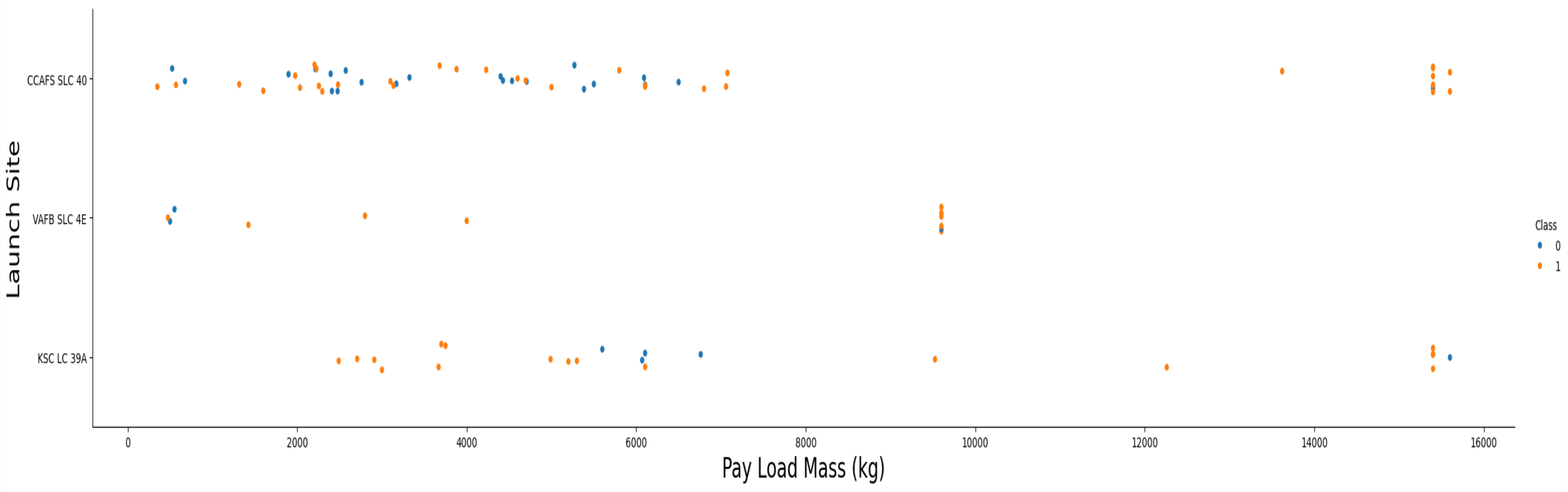
Flight Number vs. Launch Site (2/2)

- As flight number becomes larger, the occurrences of success increases, which suggests that accumulated experience in servicing, launching, payload delivery, and landing will help the success. Especially **after 80 launches, all landings were successful regardless of launch sites**.
- Most of the launch occurred in CCAFS SLC-40, which resulted in more failure in landings regardless of the flight number (*).
- There were the fewest launches from VAFB SLC-4E, and thus, there were the fewest failures in the landing. In addition, there were no launches after around the flight number 70 from the VAFB launch site (*).
- In between, there were launches from KSC LC-39A, which shows a few failures in the landing between the flight number 20 and 40 and only two failures near 80.
- The above distribution in launch sites indicates that SpaceX used the military launch sites **(CCAFS and VAFB) at the early phase** of Falcon 9 development, **then added the NASA site (KSC)** to the project.

* These observations were changed later when the relationship between the success rates and launch sites were considered. See the slide Launch Site with Higher Success Rate (2/2) in Section 3: launch Sites Proximities Analysis.

Payload vs. Launch Site (1/2)

The scatter point chart below shows the relationship between payload mass and launch site. The orange dots show successes and blue ones show failures.



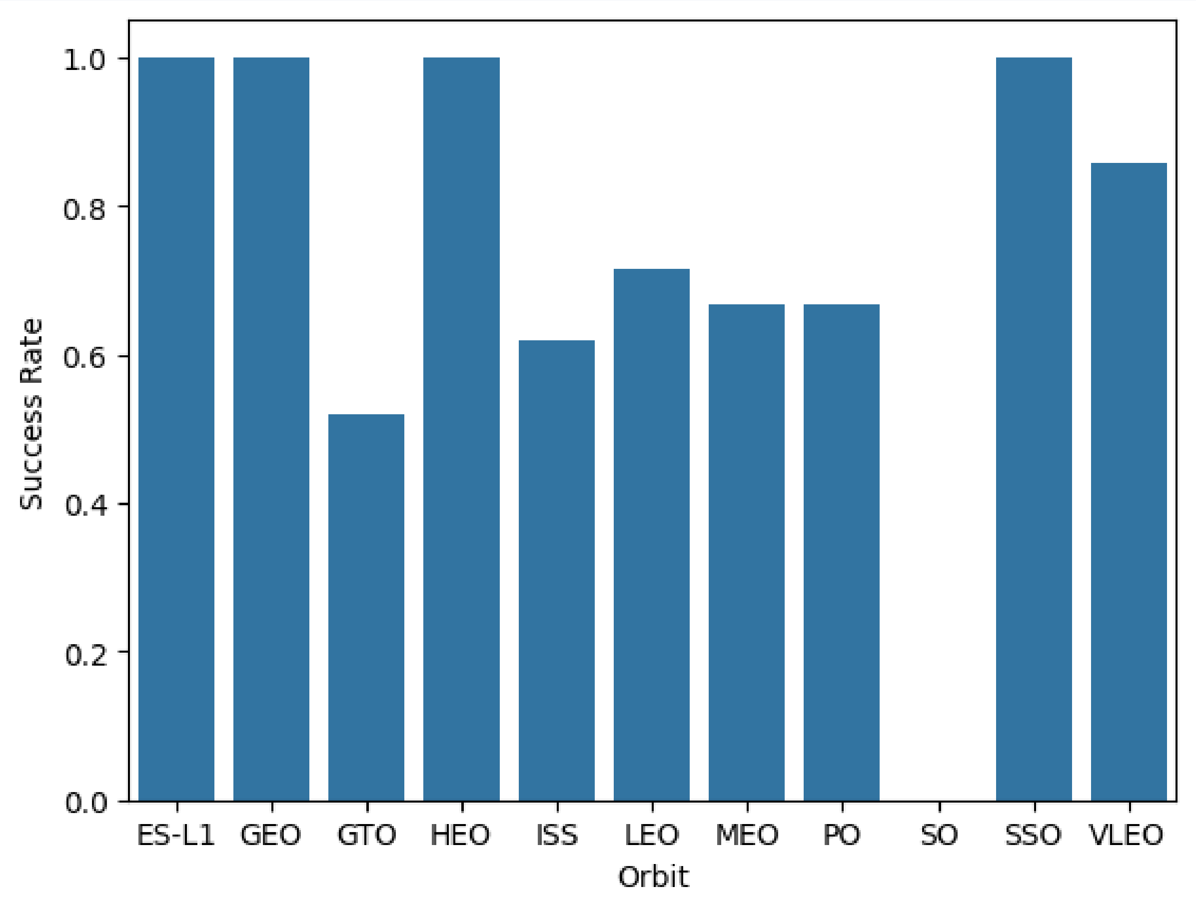
Payload vs. Launch Site (2/2)

- Most of the launches occurred with the payloads less than 8,000kg, where both successes and failures in the landing are observed:
  - ➢ Launches from CCAFS SLC-40C: a random distribution of success and failure in the landing
  - ➢ Launches from VAFB SLC-4E: more failures when the payload mass was smaller and more successes as the payload mass increased
  - ➢ Launches from KSC LC-39A: more successes in the landing when the payload mass was smaller but more failures as the payload mass increased
- The payloads between 9,000kg and 10,000kg were launched from VAFB SLC-4E and KSC LC-39A, where most of the cases were successful in the landing.
- Over 12,000kg, launches occurred in CCAFS SLC-40 and KSC LC-39A, but none in VAFB SLC-4E. In this range of the payload mass, most cases were successful in the landing, and a few failures occurred.
- The above distribution also suggests that **CCAFS SLC-40 and KSC LC-39A may have larger servicing capabilities** for heavier launches than VAFB SLC-4E does.

Success Rate vs. Orbit Type

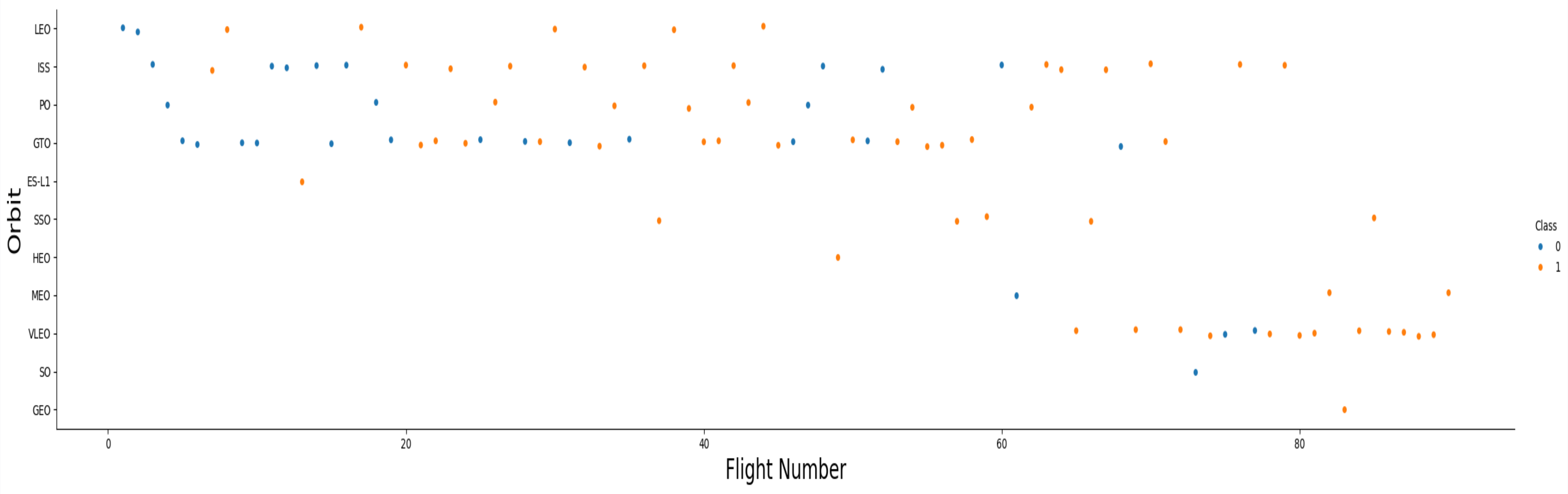
The bar chart below shows the relationship between success rate and orbit type.

- The launches to the **Earth-Sun Lagrange Point 1 (ES-L1), Geostationary Orbit (GEO), Highly Elliptical Orbit (HEO), and Sun-Synchronous Orbit (SSO)** were **all successful** in the (launching and) landing.
- The launches to the Very Low Earth Orbit (VLEO) had the success rate over 80% but not 100% in the landing.
- The launches to the International Space Station (ISS), Low Earth Orbit (LEO), Medium Earth Orbit (MEO), and Polar Orbit (PO) had the success rate over 60% but less than 80% in the landing.
- The launches to the Geostationary Transfer Orbit (GTO) had around 50% success rate but less than 60% in the landing.
- The launches to the Synchronous Orbit (SO) were not successful in the landing at all.
- For details of each orbit type, see Appendix 3.



Flight Number vs. Orbit Type (1/2)

The scatter point chart below shows the relationship between flight number and orbit type. Orange dots are success in the Falcon 9 first-stage landing whereas blue ones are failures.

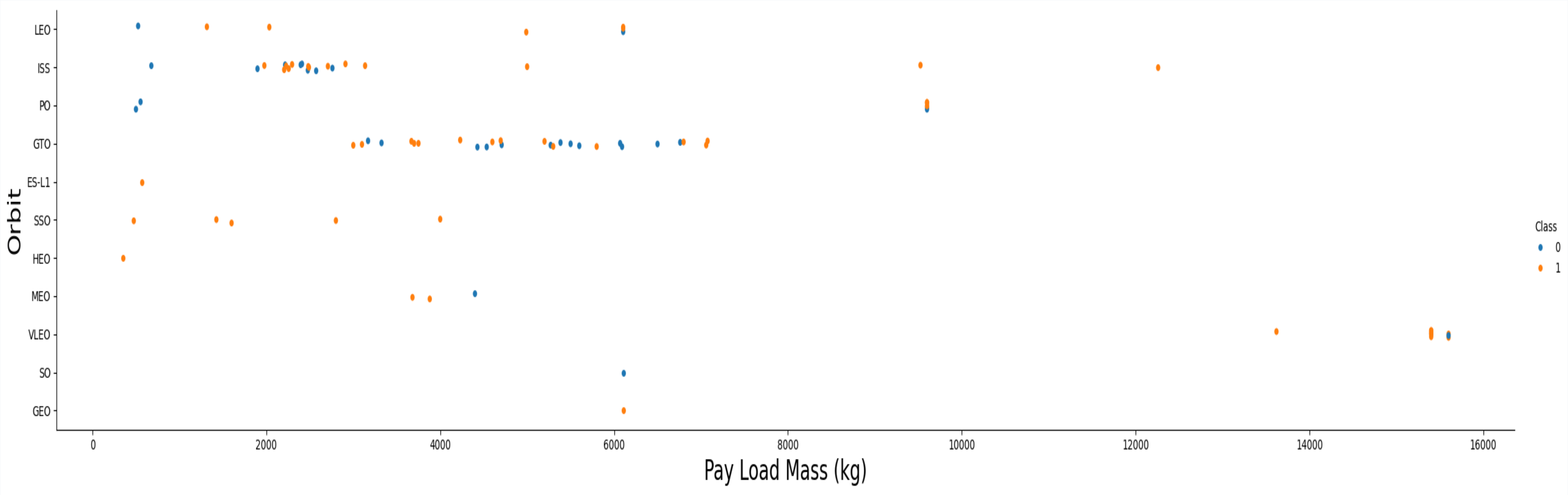


Flight Number vs. Orbit Type (2/2)

- There is a clear correlation between flight number and the orbit type LEO, which shows more successful landings as the flight number increased. However, after over 40 launches, there was no launch to the LEO.
- On the other hand, no correlation exists between the flight number and launches to the SSO, all of which were successful in the landing regardless of the flight number.
- **After the flight number 80, all launches were to the SSO, MEO, VLEO, and GEO**, and **all of their landings were successful** regardless of the orbit type.

Payload vs. Orbit Type (1/2)

The scatter point chart below shows the relationship between payload mass and orbit type. Orange dots are success in the Falcon 9 first-stage landing whereas blue ones are failures.



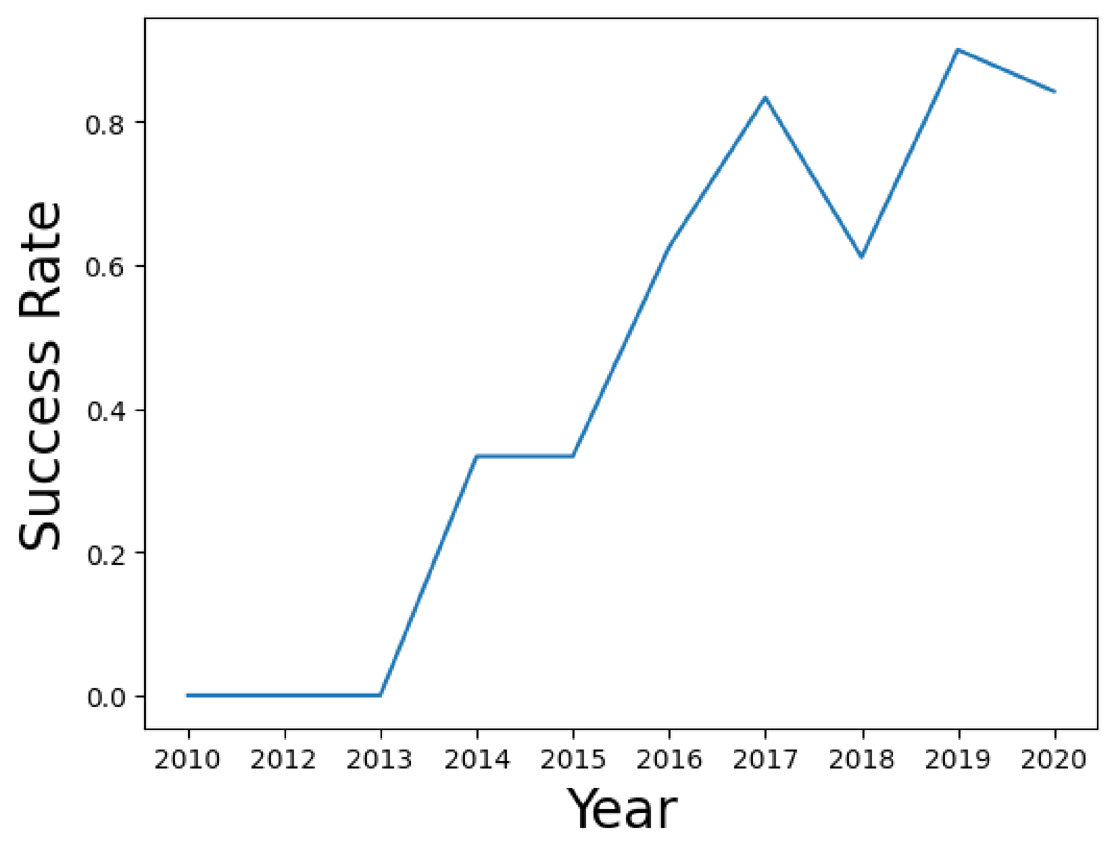
Payload vs. Orbit Type (2/2)

- The observed correlations between the payload mass and orbit type in the ranges of less than around 8,000kg, from 9,000kg to 10,000kg, and over 12,000kg respectively are like as follows.
- For payloads **less than around 8,000kg**: the chart shows **a mixed record** of success and failure in the Falcon 9 first-stage landing when the launches occurred to **the LEO, ISS, GTO, and HEO**. Launches to **the SSO were all successful** in the landing in this range of payload mass. Launches to **the EL-S1, HEO, and GEO were all successes** in the landing although the occurrences was one or just a few launches. Launches to the PO and SO were all unsuccessful in the landing with only one launch occurred to the SO.
- From **9,000kg to 10,000kg**: the chart shows that launches occurred to **the ISS and PO** and **most of the landings was successful**.
- Over **12,000kg**: the launches occurred to **the ISS and VLEO**, and the failure in the landing was observed in the launch(es) to the VLEO although **most of the landings was successful**.

Landing Success Yearly Trend

The line chart below shows the average success rate in the Falcon 9 first-stage landing year by year.

- The average success rate started rising in 2013.
- The rate leveled off in 2014 but resumed rising in 2015.
- The growth in the success rate continued until 2017 then kinked in 2018 but rose again in 2019, and then slightly dropped again in 2020.



All Launch Site Names

- SpaceX used the following launch sites for Falcon 9 launches:

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

  - In the record, there seems to be two different CCAFS launch sites, however, these are just two different launch complexes used for Falcon 9 in the same site, which initially started as LC-40 and then was transformed into SLC-40 by SpaceX (See Appendix 5 for the reference).

- The sqlight query to get the above result was %sql select distinct "LAUNCH_SITE" from SPACEXTBL;

Launch Site Names that Begin with 'KSC'

- The first 5 records where the launch sites' names start with 'KSC':

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2017-02-19	14:39:00	F9 FT B1031.1	KSC LC-39A	SpaceX CRS-10	2490	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)
2017-03-16	6:00:00	F9 FT B1030	KSC LC-39A	EchoStar 23	5600	GTO	EchoStar	Success	No attempt
2017-03-30	22:27:00	F9 FT B1021.2	KSC LC-39A	SES-10	5300	GTO	SES	Success	Success (drone ship)
2017-05-01	11:15:00	F9 FT B1032.1	KSC LC-39A	NROL-76	5300	LEO	NRO	Success	Success (ground pad)
2017-05-15	23:21:00	F9 FT B1034	KSC LC-39A	Inmarsat-5 F4	6070	GTO	Inmarsat	Success	No attempt

  - ➢ The KSC LC-39A appeared in 2017 first time in the data, which supports the previous observation on the slide "Flight Number vs. Launch Site (2/2)" that SpaceX used the military launch sites in the early phase of Falcon 9 development.
  - ➢ In addition, the appearance of KSC might be helpful for the growth in landing successes until 2017 on the slice "Landing Success Yearly Trend".

- The sqlight query to get the above result was %sql select * from SPACEXTBL where "LAUNCH_SITE" like 'KSC%' limit 5;

Total Payload Mass Carried for NASA

- Falcon 9 carried the total payload mass of 45,596kg for NASA Commercial Resupply Services (CRS), which delivered cargo and supplies to the ISS under the commercial contract.

Customer	sum(PAYLOAD_MASS__KG_)
NASA (CRS)	45596

- The sqlight query to get the above result was %sql select Customer, sum("PAYLOAD_MASS__KG_") from SPACEXTBL where Customer = 'NASA (CRS)';

Average Payload Mass Carried by F9 v1.1

- Falcon 9 booster version 1.1 (F9 v1.1) carried the average payload mass of 2,928.4kg as below:

Booster_Version	avg(PAYLOAD_MASS__KG_)
F9 v1.1	2928.4

- The above result was obtained by a sqlight query %sql select "Booster_Version", avg("PAYLOAD_MASS__KG_") from SPACEXTBL where "Booster_Version" = 'F9 v1.1';

First Successful Landing Date on a Drone Ship

- The first successful landing on a drone ship occurred on April 8th, 2016:

Landing_Outcome	min(Date)
Success (drone ship)	2016-04-08

- The above result was obtained by a sqlight query %sql select "Landing_Outcome", min(Date) from SPACEXTBL where "Landing_Outcome" = 'Success (drone ship)';

Successful Ground Landing with Payload between 4,000kg and 6,000kg

- The following Falcon 9 boosters successfully landed on ground pads with the payload mass greater than 4,000kg but less than 6,000kg:

Booster_Version
F9 FT B1032.1
F9 B4 B1040.1
F9 B4 B1043.1

- The above result was obtained by a sqlight query %sql select "Booster_Version" from SPACEXTBL where "Landing_Outcome" = 'Success (ground pad)' and "PAYLOAD_MASS__KG_" between 4001 and 5999;

Total Number of Successful and Failing Mission Outcomes

- There were 100 successes and 1 failure in mission outcomes launched by Falcon 9, although there was one case among the successes in which the payload status was unclear:

Cleaned_Outcome	Outcome
Failure (in flight)	1
Success	99
Success (payload status unclear)	1

- The above result was obtained by a sqlight query %sql SELECT TRIM("Mission_Outcome") AS Cleaned_Outcome, COUNT(*) AS Outcome FROM SPACEXTBL GROUP BY TRIM("Mission_Outcome");

Boosters that Carried the Maximum Payload

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

- The Falcon 9 booster versions in the table carried the maximum payload mass to the orbit.
- This result was obtained by a sqlight query %sql select "Booster_Version" as "Booster_Version" from SPACEXTBL where "PAYLOAD_MASS__KG_" = (select max("PAYLOAD_MASS__KG_") from SPACEXTBL);

Records of the Successful Landing in 2017

- The successful landings on ground pads in 2017 are shown below with the months, booster versions, and launch sites:

month	Date	Booster_Version	Launch_Site	Landing_Outcome
02	2017-02-19	F9 FT B1031.1	KSC LC-39A	Success (ground pad)
05	2017-05-01	F9 FT B1032.1	KSC LC-39A	Success (ground pad)
06	2017-06-03	F9 FT B1035.1	KSC LC-39A	Success (ground pad)
08	2017-08-14	F9 B4 B1039.1	KSC LC-39A	Success (ground pad)
09	2017-09-07	F9 B4 B1040.1	KSC LC-39A	Success (ground pad)
12	2017-12-15	F9 FT B1035.2	CCAFS SLC-40	Success (ground pad)

- The above result was obtained by a sqlight query %sql select substr(Date,6,2) as month, Date, "Booster_Version", "Launch_Site", "Landing_Outcome" from SPACEXTBL where "Landing_Outcome" = 'Success (ground pad)' and substr(Date,0,5)='2017';

Landing Outcomes Between 2010-06-04 and 2017-03-20

Landing_Outcome	Count_of_Outcomes
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

- The table shows the landing outcomes, e.g., failure (drone ship) or success (ground pad), that occurred between June 4th, 2010 and March 20th, 2017 in descending order.

- The result was obtained by a sqlight query %sql select "Landing_Outcome", count(*) as Count_of_Outcomes from SPACEXTBL where Date between '2010-06-04' and '2017-03-20' group by "Landing_Outcome" order by Count_of_Outcomes Desc;

Section 3

Launch Sites Proximities Analysis

Launch Site Locations (1/2)

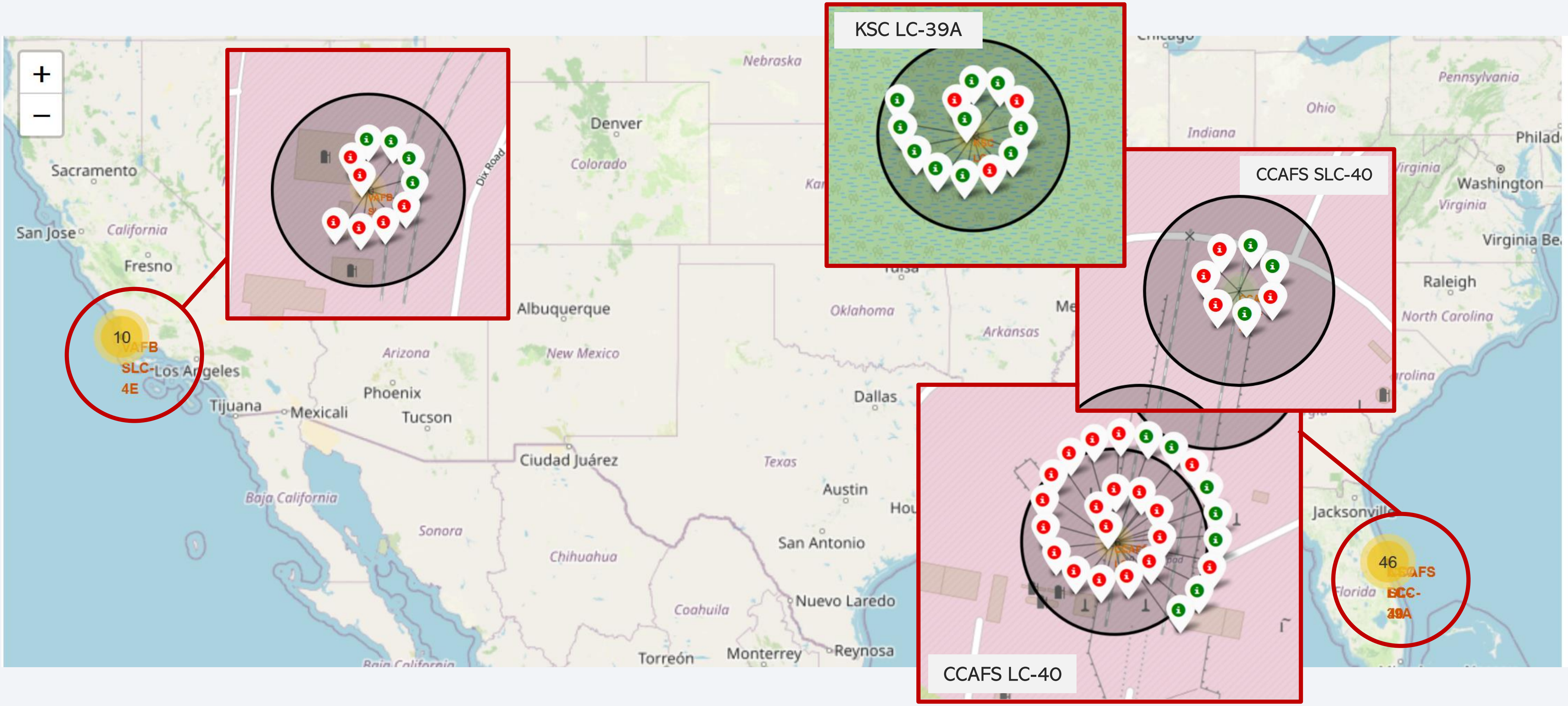
This map shows the locations of each launch site, 1 in California and 2 in Florida*.

Launch Site Locations (2/2)

- VAFB SLC-4E is located in California at **34-35 degrees** in latitude, whereas CCAFS LC-40, CCAFS SLC-40, and KSC LC-39A are located in Florida at **28-29 degrees** in latitude. The locations are comparatively **closer to the equator** in the mainland US, which provides the following benefits:
  - ➢ Extra velocity to launchers because of the Earth's rotation (only when launching eastward),
  - ➢ **Saving in fuel** for the launchers and payloads (same as above), and
  - ➢ **Easier access to the GEO**.
- The three sites in Florida are closely located with each other highly likely for the above three reasons and the already existing infrastructure since 1960s.
- When launching **to the orbit with higher inclinations than the GEO**, such as the PO and SSO, **VAFB SLC-4E has a comparative advantage** by launching northward/southward because of its latitude.
- In addition, all launch sites exist in close proximity to the coast for safety reasons. From VAFB SLC-4E, **launching southward to the open water avoids the populated area**, and from the launch sites in Florida, **launching eastward** does the same.
- In short, the launch sites both on the west and east coast nearby the equator and coastline help to attain **both efficiency and safety** in space missions.

Launch Site with Higher Success Rate (1/2)

The map below shows the occurrences of success and failure in the landing per launch site.



Launch Site with Higher Success Rate (2/2)

- Launches from **KSC LC-39A** have **the highest success rate in Falcon 9 first-stage landing**, which is 76.9% (10 successes out of 13 launches).

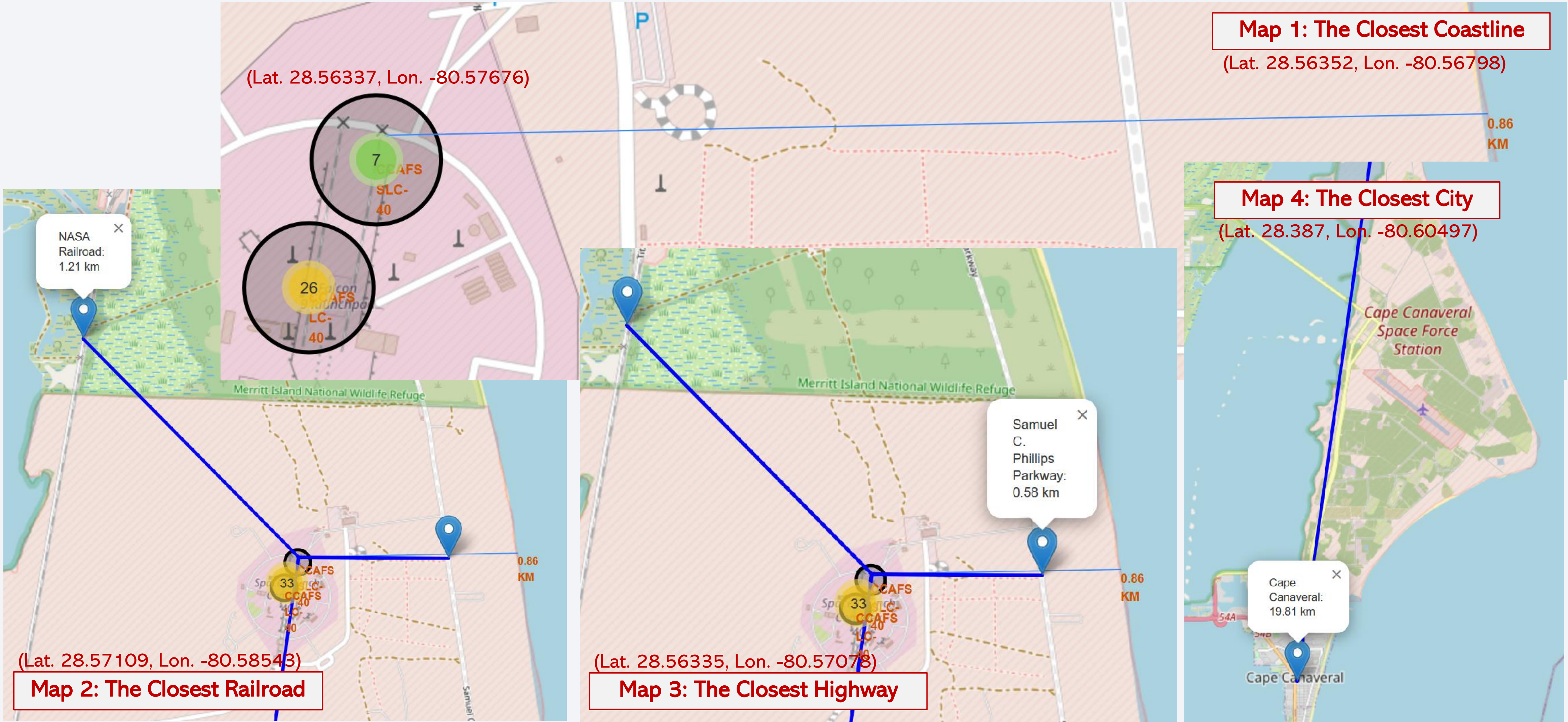
- The distribution of the success rate in descending order is:
  - KSC LC-39A: 76.9% (10 successes out of 13 launches)
  - CCAFS SLC-40: 42.9% (3 successes out of 7 launches)
  - VAFB SLC-4E: 40% (4 successes out of 10 launches)
  - CCAFS LC-40: 26.9% (7 successes out of 26 launches)

- This distribution **betrays the observation** on the slide “Flight Number vs. Launch Site (2/2)” that there is **a correlation between the number of launches from each site and cases of failure**, even when launches from the CCAFS are combined. When the observation is checked with the rate of success/failure, the VAFB shows the higher failure rate than the CCAFS.

- The above finding suggests that for the success of Falcon 9 first-stage landing, other factors than the number of launches have to be taken into consideration, such as the target orbit, payload, and probably the match between mission objectives and servicing capabilities of the launch sites.

Proximities of CCAFS SLC-40 (1/2)

The map below shows the distance to the closest coastline, railroad, highway, and city from the CCAFS SLC-40.



Proximities of CCAFS SLC-40 (2/2)

- **The coastline (Map 1)**: the CCAFS SLC-40 is located in the close proximity of the coastline, which is about **0.86km** from the site. This is better for **safety reasons** than otherwise. For example, disposable components of launchers will be dropped over the ocean instead of populated areas, and in the worst case such as self destruct/explosion of a launcher, the risk to people may be minimized if it happens over the ocean.

- **The closest railroad (Map 2)**: NASA Railroad is located at about **1.21km** from the site. It appears that the railroad is used to transport huge components/subsystems for launchers. However, the rail stopped at 1.21km north to the site, and the rail is connected to Titan III Road. Probably the cargo was trans-shipped to tracks/trailers and delivered to the site.

- **The closest highway (Map 3)**: Samuel C. Phillips Parkway is located at about **0.58km** from the site. The highway goes through the Cape and eventually reached residential areas in the city of Cape Canaveral. This highway appears to be used by employees, contractors, and visitors on business to commute to the site as well as for tourists to visit the space complex in the area.

- **The closest city (Map 4)**: Cape Canaveral lies at about **19.81km** from the site. The city is **distant from the site** compared to the railroad, highway, and the coastline. This is rational for the safety reasons. At the same time, about 20km is a commutable distance, so probably many employees and contractors of the site live in this city.

Section 4

Build a Dashboard with Plotly Dash

Launch Success Distribution among Launch Sites

The dashboard below shows the distribution of launch success across the launch sites.

All Sites

Total Success Launches by Site

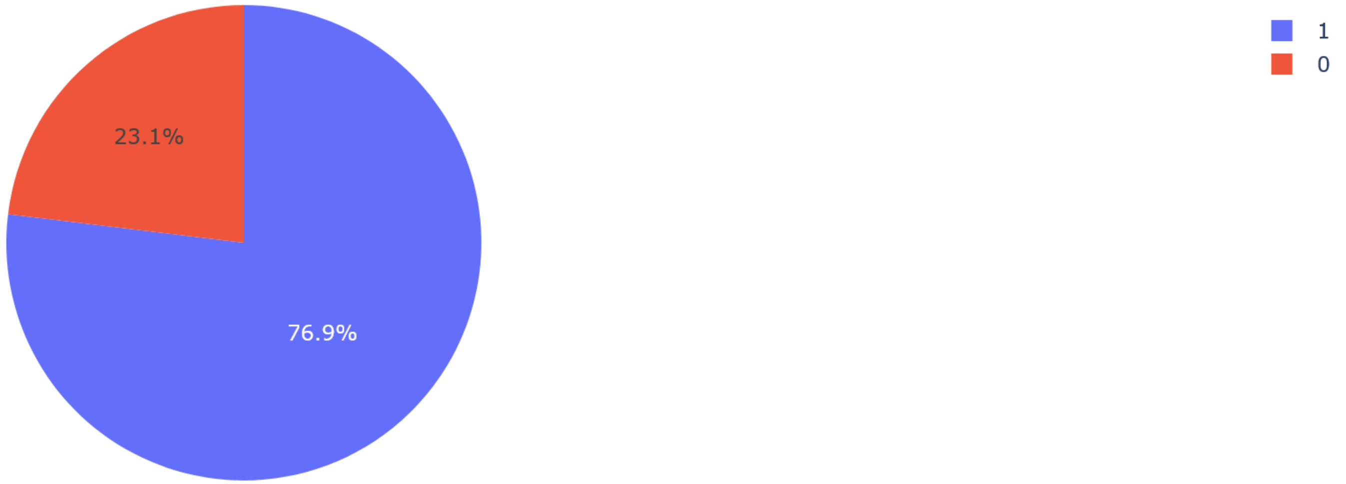
- KSC LC-39A has the largest successful launches, i.e., Falcon 9 first-stage landings.
- CCAFS LC-40 follows KSC, then comes VAFS SLC-4E.
- CCAFS SLC-40 has the fewest success in the launches and landings.

Launch Site with the Highest Launch Success Ratio

The dashboard below shows the launch success ratio of KSC LC-39A.

KSC LC-39A

Total Success Launches for site KSC LC-39A



- KSC LC-39A also has the highest launch success rate, which was found by:
  - Select each site in the dropdown menu and then pointing the shown pie chart to find the counts of success and failure of launches from the site. Then calculate the rate as follows.

{[Success Launch Counts per Site of KSC LC-39A, CCAFS SLC-40, VAFB SLC-4E, CCAFS LC-40 (10, 3, 4, 7)]/Total Launch Counts (Success + Failure) per Site (13, 7, 10, 26)} X 100(%) = (76.9%, 42.9%, 40%, 26.9%)

Payload vs. Launch Outcome for All Sites (1/3)

The dashboard below shows the launch outcome per range of payload mass.

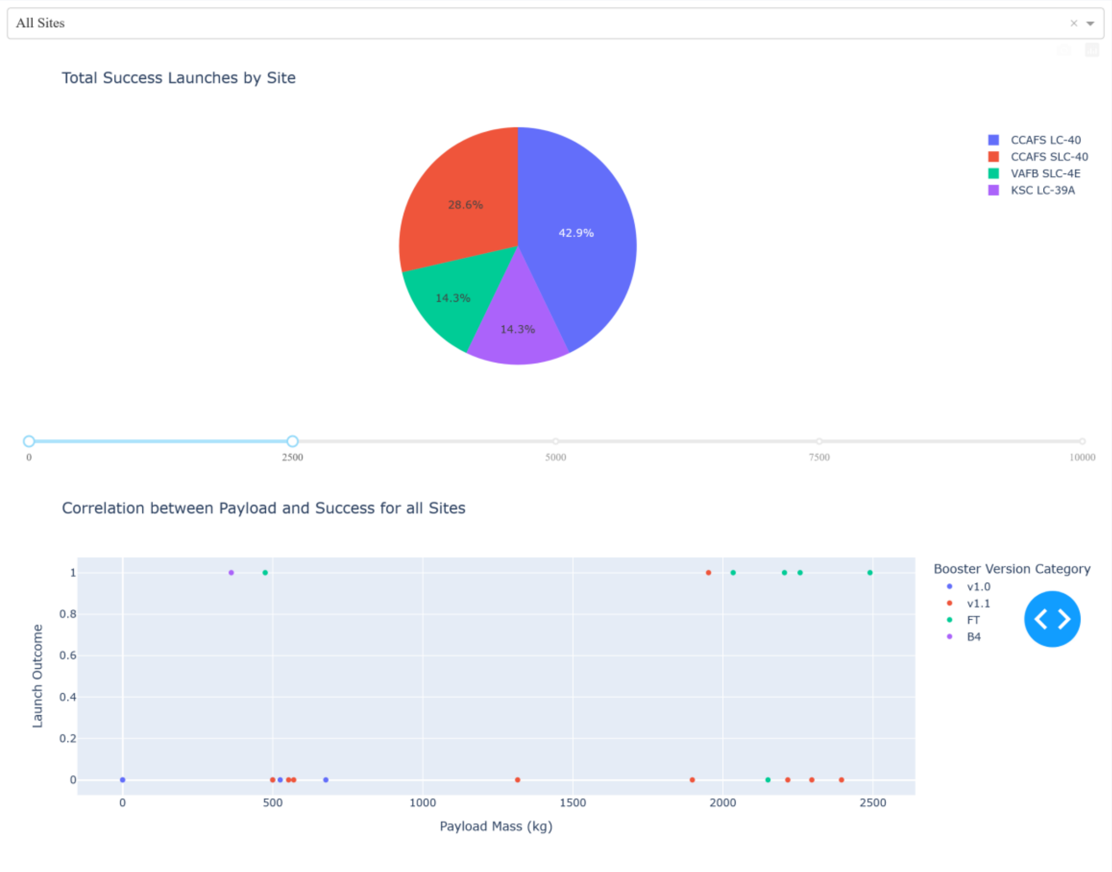


Figure 1: 0kg to 2,500kg

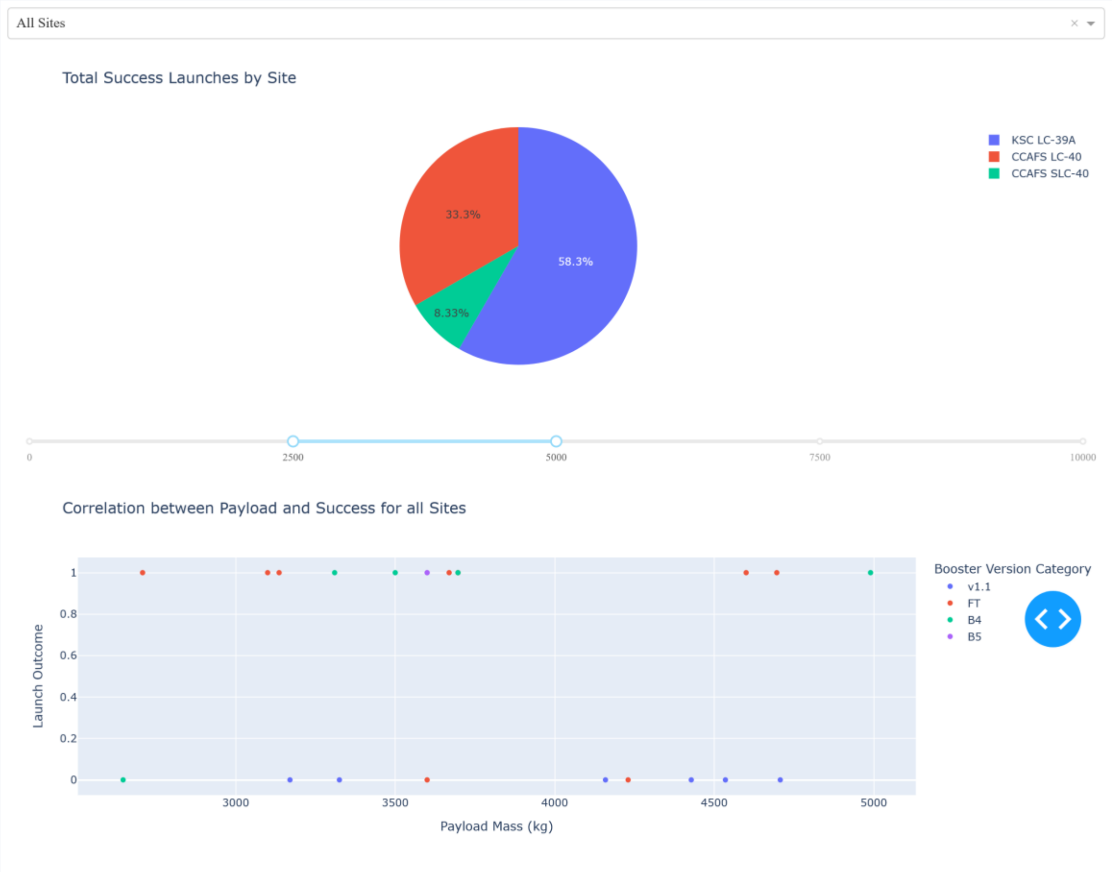


Figure 2: 2,500kg to 5,000kg

Payload vs. Launch Outcome for All Sites (2/3)

The dashboard below shows the launch outcome per range of payload mass.

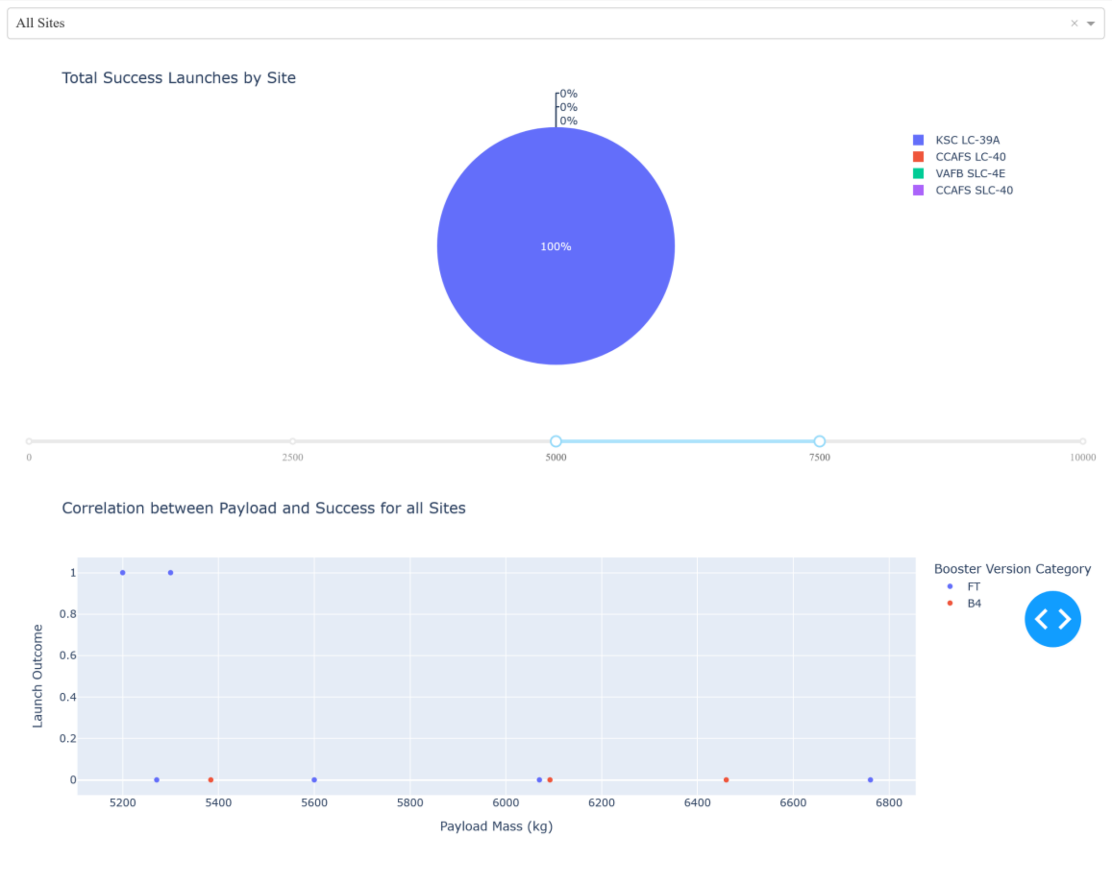


Figure 3: 5,000kg to 7,500kg

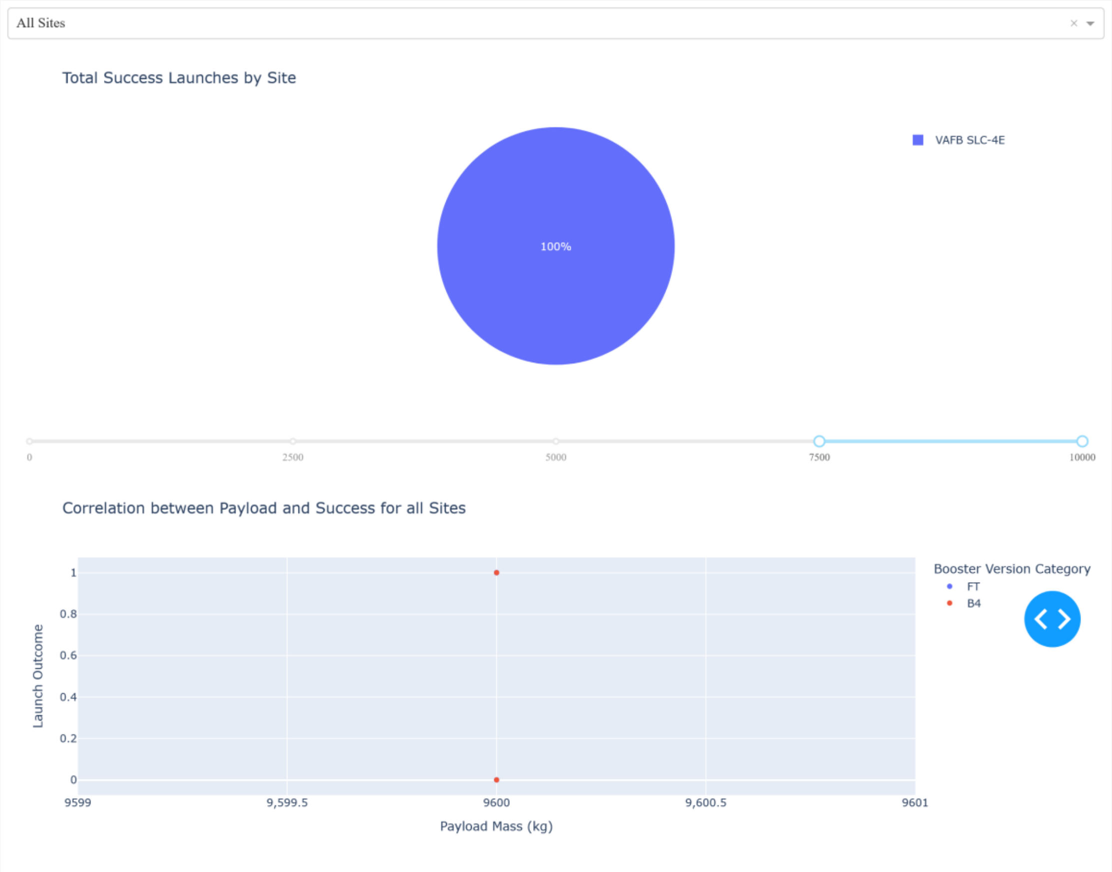


Figure 2: 7.500kg to 10,000kg

Payload vs. Launch Outcome for All Sites (3/3)

- The success rate per payload mass range
  - ➢ The range from **2,500kg to 5,000kg** shows **the highest success rate**.
  - ➢ The range from 5,000kg to 7,500kg shows the lowest success rate.
  - ➢ In addition, in the range between 5,000 and 7,500kg, only KSC LC-39A has the success in the launches (counts=2), and in the range from 7,500 to 10,000kg, only the VAFB SLC-4E has the successful launches (counts = 3).
- The Falcon 9 booster version (v1.0, v1.1, FT, B4, B5, etc.)
  - ➢ The booster version **B5** has the highest success rate.
- The range should be 0-2,499kg, for example, for mathematical accuracy, but in this case the current description does not make a big difference.
- Appendix 4 describes how the above values were obtained using the dropdown menu.

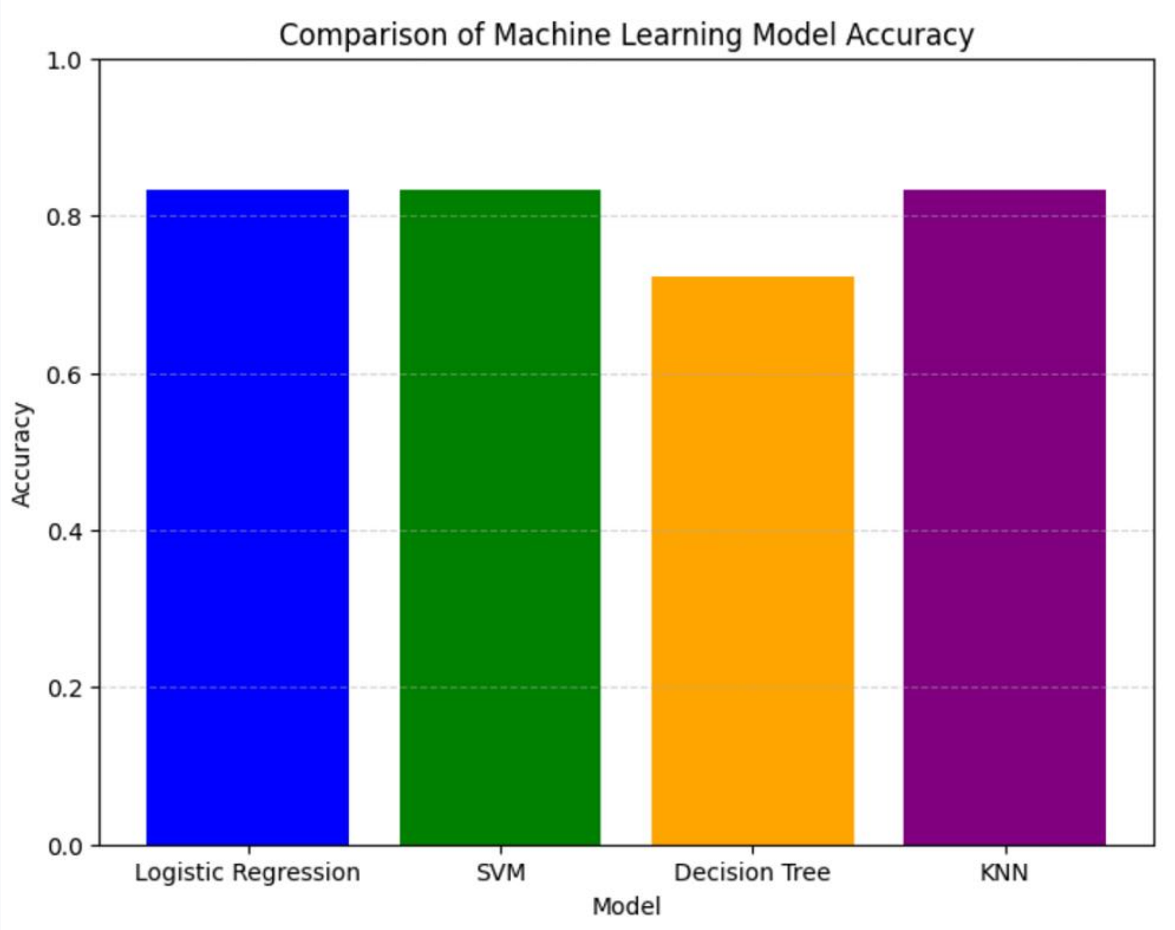
Section 5

Predictive Analysis (Classification)

Classification Accuracy of Machine Learning Models

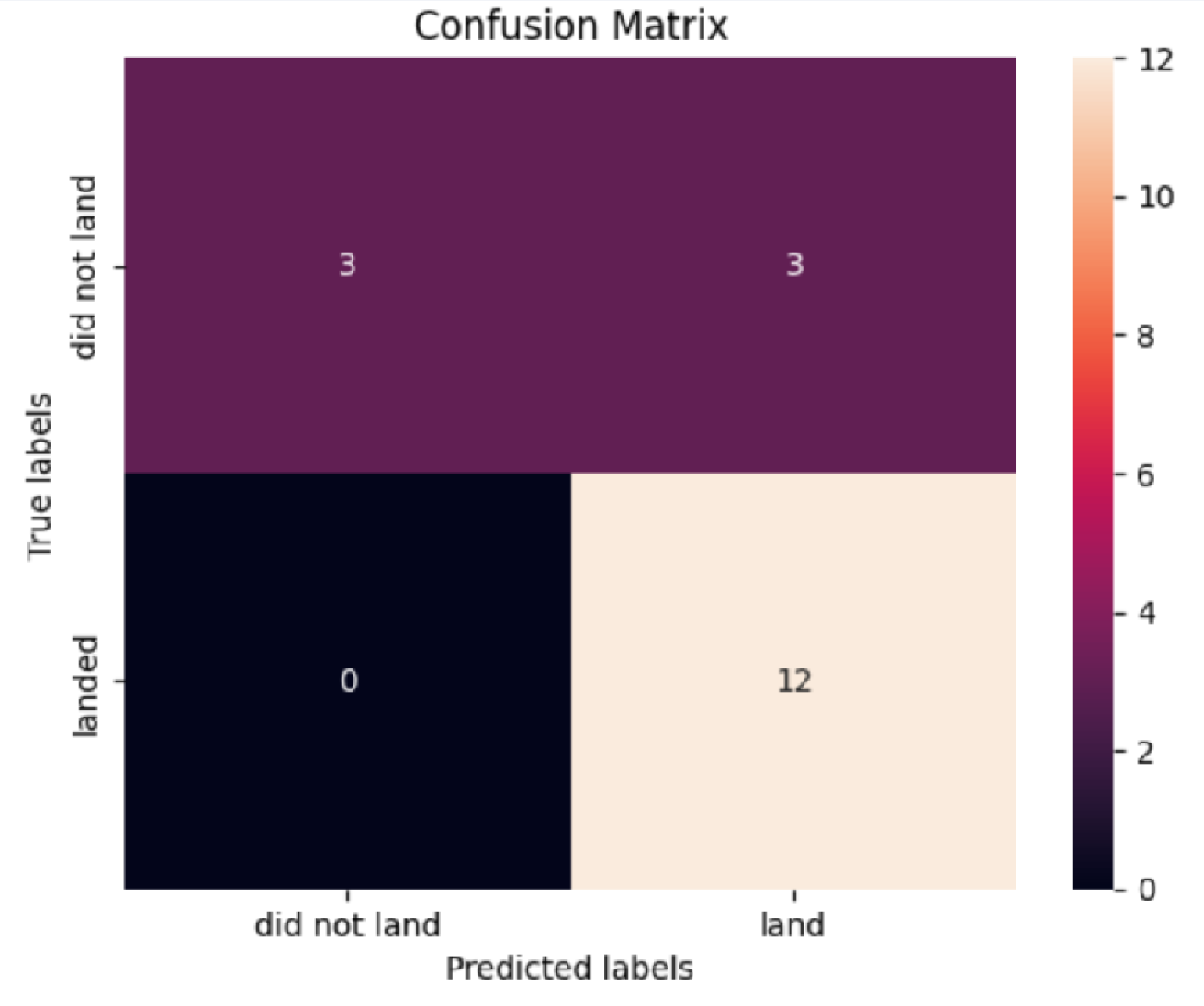
The bar chart compares accuracy of machine learning models to predict whether Falcon 9 first-stage successfully lands with the given dataset.

- The decision tree has a lower accuracy, but other three models, Logistic Regression, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN), show the same power for the prediction.
- With the three models that have the same accuracy, the **logistic regression** is recommended because of its **interpretability** (having coefficients), **training time** (faster than SVM and KNN), and **scalability** (the second to the best in this regard may be SVM).
- When it comes to robustness to overfitting, SVM may be the best, but there are the other three factors in this case mentioned above.



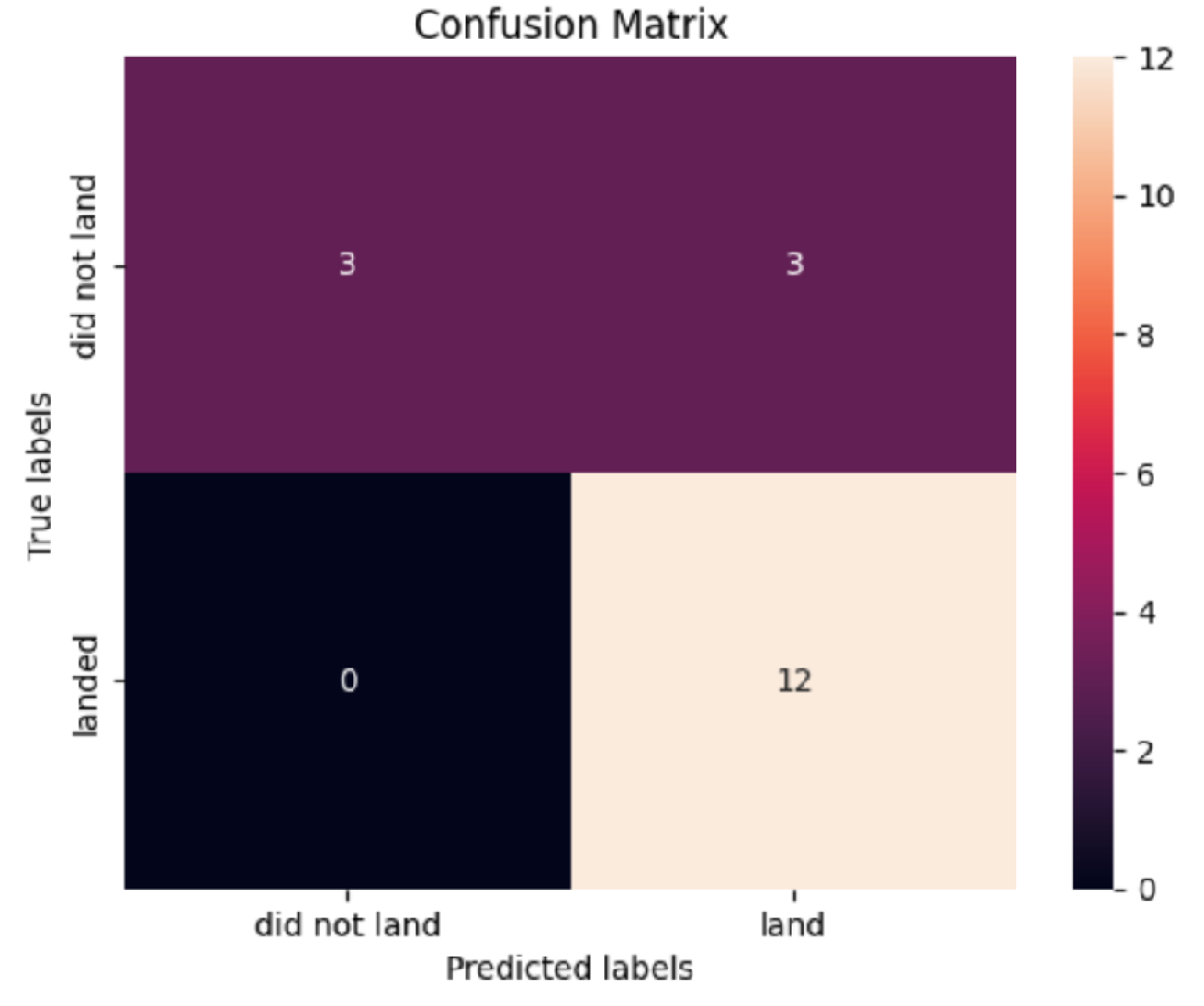
Confusion Matrix of Logistic Regression Model (1/2)

- The confusion matrix illustrates the performance of logistic regression model to predict successful Falcon 9 first-stage landing.
- The breakdown of correct and incorrect predictions is as follows:
  - ➢ True Positives: 15 predicted successful landings for 12 successful landings
  - ➢ True Negatives: 3 predicted failures out of 3 unsuccessful landings
  - ➢ False Positives: 3 predicted successful landings where there are no such successes.
  - ➢ False Negatives: 0 predicted successful landings where there are no success.



Confusion Matrix of Logistic Regression Model (2/2)

- The logistic regression model is reliable for predicting failures in Falcon 9 first-stage landing:
  - ➢ False positives = 0
  - ➢ True negatives = 0
- Predictive power of successful landing needs to be improved:
  - ➢ False positives = 3, which will be a serious concern in the prediction and further business analytics
- How to improve the accuracy:
  - ➢ For the purpose of this project, the size of the dataset was limited, so accumulating cases of Falcon 9 launches and landings will contribute to improve the accuracy of the model.



Conclusion (1/2)

As far as the available data for this project suggest, the following points can be said as the conclusion.

- **The development of Falcon 9 for its first-stage landing progressed well**, which is good news for potential customers to reduce the launch cost.
  - After the Flight Number 80, all landings were successful.
  - The landing sites added their variety: the first landing success on a drone ship was in 2016 and on a ground pad was in 2017.
  - The customer base was broadened: service to the NASA CRS started in 2017.
  - The booster for Falcon 9 was continually upgraded from v1.1 (the average payload delivery capability 2,928.4kg), then FT and B4 (> 4,000kg and < 6,000kg), and finally B5 with the largest capability and highest success rate in the first-stage landing.
  - The mission success rate was also high: over 99 out of 100 launches.
- To predict whether the Falcon 9 first-stage lands successfully or not, a machine learning model of **logistic regression** is recommended.
  - Accuracy is over 0.8 and still needs improvement: the model is reliable to predict failures in the landing, but its predictive power of successful landing (false positives = 3) is a concern for prediction and business analytics.

Conclusion (2/2)

- The launch cost reduction by the reusable Falcon 9 first-stage is very positive to space exploration and utilization, but **how data science will help you** in winning the space race depends on whether you are **a customer or competitor of SpaceX in the market**.
    - ➢ For a customer, successful payload delivery matters most for which the launch is part of the total cost (although the launch clearly accounts for its huge part) . For example, the orbit type depends on the payload, and this condition as well as the payload mass influence the choice of the launch site.
    - ➢ In this regard, the analysis in this project about the launch site, payload mass, and orbit type is as much helpful as the machine learning for predicting the successful first-stage landing.
    - ➢ For a competitor in the launch service market, the launch cost itself is the key to the strategic positioning. In this case, the machine learning of this project is the primary concern in addition to the analysis about landing conditions of Falcon 9 first-stage. Then, the analysis of other conditions concerning payload delivery adds values to business decision-making.

Appendices

- Appendix 1: SpaceX Dataset by Data Wrangling
- Appendix 2: Launch Site Details
- Appendix 3: Orbit Types
- Appendix 4: The Dropdown Menu and Success/Failure Rates
- Appendix 5: References

Appendix 1: SpaceX Dataset by Data Wrangling (1/3)

1. Missing Values:

   About 29% of the cells in ‘LandingPad’ was missing, but this was planned in the development process of Falcon 9, in which the landing pads were not used.

2. Data Types:

Data Type	Attributes
int64	FlightNumber, Flights, and ReusedCount
float64	PayloadMass, Block, Longitude, and Latitude
object	Date, BoosterVersion, Orbit, LaunchSite, Outcome, LandingPad, and Serial
boolean	GridFins, Reused, and Legs

3. Launches from Each Launch Site:

Launch Site	Launches
Cape Canaveral Space Launch Complex 40 (CCAFS SLC-40)	55 launches
Kennedy Space Center Launch Complex 39A (KSC LC-39A)	22 launches
Vandenberg Air Force Base Space Launch Complex 4E (VAFB SLC-4E)	13 launches

Appendix 1: SpaceX Dataset by Data Wrangling (2/3)

4. Occurrences to the Orbits

Orbit	Launches
Geostationary Transfer Orbit (GTO)	27 launches
International Space Station (ISS)	21 launches
Very Low Earth Orbits (VLEO)	14 launches
Polar Orbit (PO)	9 launches
Low Earth Orbit (LEO)	7 launches
Sun-Synchronous Orbit (SSO)	5 launches
Synchronous Orbit (SO)	1 launch
Medium Earth Orbit (MEO)	3 launches
Earth-Sun System Lagrangian Point #1 (ES-L1)	1 launch
Highly Elliptical Orbit (HEO)	1 launch
Geostationary Orbit (GEO)	1 launch

Appendix 1: SpaceX Dataset by Data Wrangling (3/3)

5. Occurrences of Mission Outcome

Mission Outcome	Launches
Successfully Landed on a Drone Ship (True ASDS)	41 launches
Failed in Landing on a Drone Ship (False ASDS)	6 launches
Successfully Landed to a Specific Region of the Ocean (True Ocean)	5 launches
Failed in Landing to the Ocean (False Ocean)	2 launches
Successfully Landed on a Ground Pad (True RTLS)	14 launches
Failed in Landing on a Ground Pad (False RTLS)	1 launches
Landing Itself Failed (None ASDS and None None)	21 launches

Appendix 2: Launch Site Details

Launch Site	Location	Owner	Tenant
Kennedy Space Center Launch Complex 39A (KSC LC-39A)	Merritt Island, Florida (28.6083°N, 80.6044°W)	National Aeronautics and Space Administration (NASA)	SpaceX
Cape Canaveral Space Launch Complex 40 (CCAFS LC-40, SLC-40)	Cape Canaveral Space Force Station, Florida (28.56194°N, 80.57722°W, 28.5621°N, 80.5772°W)	United States Space Force	SpaceX
Vandenberg Space Launch Complex 4 East (VAFB SLC-4E)	Vandenberg Space Force Base, California (34.633°N, 120.613°W)	United States Space Force	SpaceX

Appendix 3: Orbit Types

Orbit	Altitude (km)	Eccentricity*	Inclination (deg.)	Note
ES-L1 (Earth-Sun Lagrange Point 1)	~1,500,000	Low to moderate	~0	A three-dimensional orbit created by the Sun, Earth, and satellite in which a small object (satellite) is in equilibrium to the center of mass of larger objects (the Sun and Earth). Used for space telescope, solar observation, and deep-space communication, etc.
Geostationary Orbit (GEO) **	35,786	Nearly zero	~0	Valuable for the weather, communication, and surveillance satellites
Geostationary Transfer Orbit (GTO)	~200 - 35,786	High	~27	Used to transfer a satellite to GEO. When launched from KSC/CCAFS, the inclination is around 27 degrees
Highly Elliptical Orbit (HEO)	500 - 40,000	Varies	Varies	An elliptical orbit with high eccentricity
Low Earth Orbit (LEO)	=<2,000	<0.25 (near circular)	0 - 90	The orbit of the ISS
Medium Earth Orbit (MEO)	2,000 - 35,786	Low to moderate	0 - 90	Also called an intermediate circular orbit when it is at 20,200km or 20.650km which makes an orbital period 12 hours
Polar Orbit (PO)	200 - 1,000	Low to moderate	Close to 90	The orbit that passes over two poles with which a satellite covers the entire planet because of the Earth's rotation
Sun-Synchronous (heliosynchronous) Orbit (SSO)	600 - 800	Low to moderate	~98	A nearly polar orbit for a satellite to cover a fixed point on the planetary surface at the same local mean time
Synchronous Orbit (SO)	Varies	Varies	Varies	Can be equatorial and circular, or non-equatorial and elliptical
Very Low Earth Orbit (VLEO)	<450	Low	0 - 90	Beneficial to earth observation

* Eccentricity: an index about the orbit shape whose value takes between 0 and 1 where 0 means circular. As the value increases, the shape becomes more elliptical.
** If a satellite has an inclination larger than 0 at the altitude of 35,786km, the orbit is called Geosynchronous Orbit (GSO).

Appendix 4: The Dropdown Menu and Success/Failure Rates

- The success rate per payload mass range
  - ➢ Selecting all sites in the dropdown menu, sliding the range on the slider, and then pointing the shown pie chart gives the following:

  {[Success Launch Counts per Range of 0-2,500, 2,500-5,000, 5,000-7,500, and 7,500-10,000 in kg (7, 12, 2, 3)]/Total Success Counts (13+7+10+26 = 56)} X 100(%) = (12.5%, 21.43%, 3.57%, 5.36%)

- The Falcon 9 booster version
  - ➢ Clicking the booster version in the legend and then counting success versus failure gives the following:

  {[Success Launch Counts per Booster Version of v1.0, v1.1, FT, B4, and B5 (0, 1, 15, 6, 1)]/Total Launch Counts per Booster (4, 15, 23, 11, 1)} X 100(%) = (0%, 6.67%, 65.2%, 54.5%, 100%)

  - ➢ The total launch counts for this question was 54, not 56, missing two launch counts. This occurs because the two values of payload mass are the same, and the table cannot show the two launch separately as in the Failure of v1.0 and the Success in FT.

Appendix 5: References

- The relationship between CCAFS LC-40 and SCL-40: Accessed April 20, 2025
    - https://en.wikipedia.org/wiki/Cape_Canaveral_Space_Launch_Complex_40
- Launch Site Details: Accessed May 10, 2025
    - https://en.wikipedia.org/wiki/Cape_Canaveral_Space_Launch_Complex_40
    - https://en.wikipedia.org/wiki/Vandenberg_Space_Launch_Complex_4
    - https://latitude.to/articles-by-country/us/united-states/23847/cape-canaveral-air-force-station-space-launch-complex-40
- Orbit Types: Accessed May 10, 2025
    - Lab 2: Data wrangling of IBM DS0720EN Data Science and Machine Learning Capstone Project
    - https://en.wikipedia.org/wiki/List_of_orbits
    - https://en.wikipedia.org/wiki/Synchronous_orbit
    - https://www.esa.int/Enabling_Support/Space_Transportation/Types_of_orbits
    - https://www.space.com/29222-geosynchronous-orbit.html
    - Virgili-Llop, Josep; Peter C. E. Roberts; Daniel Zhou Hao; Laia Ramio-Tomas; Valentin Beauplet. 2014. "Very Low Earth Orbit mission concepts for Earth Observation. Benefits and challenges". *Reinventing Space Conference*, 18-21 November 2014, London, UKAt: Royal Society, London, UK. Downloaded from ResearchGate on May 10, 2025.

Thank you!